

THE g_1 PROBLEM: DEEP INELASTIC ELECTRON SCATTERING AND THE SPIN OF THE PROTON*

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A recent measurement of the spin-dependent structure function of the proton, $g_1^p(x, Q^2)$, suggests that the quark's spin accounts for little of the proton's spin. We analyze several theoretical issues bearing on this result. We discuss the result expected in a variety of quark models. We examine the proposal that gluons contribute to the proton's spin in an amount calculable (in terms of gluon distribution functions) in perturbative QCD. We examine the conserved currents associated with Lorentz transformations in QCD, derive sum rules and show that the anomalous (gluonic) $U(1)_A$ current, K_μ , is not in general to be identified with the gluon spin. Finally, we discuss the ensemble of nucleon matrix elements of quark bilinear operators and urge an experimental and theoretical program to study them.

1. Introduction

There is a widespread impression in the particle physics community that something is wrong with the spin of the proton: A recent measurement [1] by the European Muon Collaboration (EMC) indicates that only a small percentage of the proton's spin is carried by the spin angular momentum of the quarks. There are many caveats associated with this interpretation of the experiment: a dangerous extrapolation is required [2]; $SU(3)_{\text{flavor}}$ symmetry violation could alter the result significantly [3]; the mass scale of the experiment differs significantly from the scale at which constituent quark models are formulated [4]. Nevertheless, unless the experiment is simply wrong, it appears that most of the proton's spin is

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to be found on something other than the spin of its quarks. A re-analysis [5] of the results of a very different experiment, low-energy elastic neutrino–proton scattering [6, 7], supports the same conclusion.

The above result has been advertised variously as: evidence of the breakdown of QCD [8]; a reliable prediction and success of the three-flavor Skyrme model [9]; an indication of significant configuration mixing in the nucleon wavefunction [10]. Recently, several interesting papers [11–13] have examined the possibility that gluons contribute significantly to the spin of the nucleon in an amount which can be calculated using traditional methods of perturbative QCD. These papers have stimulated considerable further work [14–17].

The aim of this paper is to argue that the EMC and νp scattering results alone present no deep puzzle. They are a significant, but by no means unforeseeable, violation of the OZI rule [18]. The EMC result appears more surprising, we shall argue, when it is viewed in comparison with the success of the OZI rule in other areas of deep inelastic phenomenology, notably in the “momentum sum rule” [19, 20]. This viewpoint suggests a program of future measurements [5, 21] of the nucleon matrix elements of quark bilinear operators of various flavors [22, 23], which could shed considerable light on the flavor content of the nucleon and which should be carried out.

The aim of this paper is to investigate various theoretical issues surrounding the interpretation of the EMC result. It is not a review, although much of our work was stimulated by our attempts to understand theoretical papers motivated by the data. In sect. 2 we give a very brief review of the deep inelastic polarization asymmetry and the sum rule which relates it to the quark spin. We also mention the $\nu p \rightarrow \nu p$ experiment and its relation to the same matrix element. We briefly discuss some issues of flavor SU(3) violation, the scale dependence of operator matrix elements and the role of gluons, looking ahead to later sections of the paper. In sect. 3 we discuss the “naive” expectation for the fraction of the proton’s spin carried by the quark’s spin. The answer is not unity, but more like 0.60 ± 0.12 . In sect. 4 we briefly look at the effect of SU(3)-flavor symmetry violation on the interpretation of the EMC data. We find that SU(3) works surprisingly well for the matrix elements of interest, making it unlikely that the effect is merely a consequence of misunderstood SU(3) violation. In sect. 5 we study the problem from the point of view of the operator product expansion, and chiral U(1) Ward identities. We use this approach to analyze the recent work of Altarelli and Ross [12] and Carlitz et al. [13]. These authors suggest that the triangle anomaly gives rise to an independent gluonic contribution to the polarization sum rule which can be calculated in perturbative QCD (up to unknown gluon distribution functions):

$$\Delta q = \widetilde{\Delta q} - \frac{\alpha_s}{2\pi} N_f \Delta g, \quad (1.1)$$

where Δq is the measured integrated asymmetry and the second term on the

right-hand side is the gluonic contribution. They further argue (i) that $-(\alpha_s/2\pi)N_f \Delta g$ can be observed in two-jet production by polarized electrons from a polarized target and (ii) that $\widetilde{\Delta q}$ can be identified with the “naive” quark model expectation $\widetilde{\Delta q} = 0.60 \pm 0.12$. Thus the EMC data, $\Delta q \approx 0$, are to be understood qualitatively by a cancellation of a large $\widetilde{\Delta q}$ against a large gluonic term. We agree with point (i) (for a recent exposition see ref. [24]), but disagree with point (ii).

In sect. 5 we argue that there does not appear to be a unique way to separate out a “gluonic contribution” to Δq , and more important, if one proceeds as in refs. [12, 13] by *defining* the gluonic contribution to be $-(\alpha_s/2\pi)N_f \Delta g$ (as, in principle, measurable in two-jet production) then what remains, defined to be $\widetilde{\Delta q}$, is not the simple quark model axial charge. This is particularly clear for the case of massive quarks, which we describe in detail. We also show that the effects which spoil the simple interpretation of $\widetilde{\Delta q}$ in the case of massless quarks are easily overlooked in a perturbative calculation.

In sect. 6 we present the operator description of spin and orbital angular momentum in field theories of quarks and gluons. Although this analysis is straightforward, even elementary, there has been some recent confusion [11, 14] as to (i) whether orbital angular momentum of quarks and gluons contribute in an infinite momentum frame, and (ii) what operator measures the gluon spin in QCD. We show that quarks and gluons can have arbitrary orbital angular momentum without large $k_\perp$ in an infinite momentum frame. We find that the quark flavor singlet axial current is a direct measure of the spin component of angular momentum, but that the situation is very different for gluons. In sect. 7 we discuss calculations of the angular momentum content of the nucleon in the $N_c \rightarrow \infty$ limit in chiral soliton (Skyrme) models. We briefly describe and compare two of the calculations in the literature [9, 25]. We note that the same limit of the Skyrme model is likely to make unsuccessful predictions for the flavor structure of other bilinear quark operators. As an example we refer to the recent calculation of the matrix elements of $\bar{q}q$ by Kaplan and Klebanov [26] who find that the $N_c \rightarrow \infty$ limit of the Skyrme model gives a poor description of these matrix elements. Finally, in sect. 8 we discuss the flavor structure of the ensemble of bilinear quark operators, how they are constrained and how they might be measured.

2. Review

In 1987 the EMC [1] reported a measurement of the spin-dependent proton structure function, $g_1^p(x, Q^2)$. The quantity measured is the spin asymmetry,

$$A_1 \equiv \frac{\mu^\uparrow p^\uparrow - \mu^\uparrow p^\downarrow}{\mu^\uparrow p^\uparrow + \mu^\uparrow p^\downarrow}, \quad (2.1)$$

where the terms denote cross sections and the arrows $\uparrow$ and $\downarrow$ denote polarizations along the beam direction. A_1 is related to g_1 by

$$g_1 \cong A_1 F_1 / (1 + R), \quad (2.2)$$

up to terms which are negligible in the deep inelastic limit. F_1 and $R = \sigma_L / \sigma_T$ are the conventional unpolarized target observables in deep inelastic muon scattering.

There is a reliable, well-established QCD prediction for the lowest moment of $g_1^p(x)$ [27],

$$\begin{aligned} \int_0^1 dx g_1^p(x, Q^2) &= \frac{1}{2} \left[\frac{4}{9} \Delta u(Q^2) + \frac{1}{9} \Delta d(Q^2) + \frac{1}{9} \Delta s(Q^2) \right] \\ &\times \left(1 - \frac{\alpha_s(Q^2)}{\pi} + O(\alpha_s^2) \right) + O\left(\frac{\Lambda^2}{Q^2}\right), \end{aligned} \quad (2.3)$$

where

$$\Delta\psi(\mu^2) s_\alpha = \langle p, s | \bar{\psi} \gamma_\alpha \gamma_5 \psi |_{\mu^2} | p, s \rangle. \quad (2.4)$$

$|p, s\rangle$ is the proton of momentum p , spin s with $s^2 = -M^2$, $s \cdot p = 0$, normalized to $\langle p, s | p', s \rangle = 2E(2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}')$. The label μ refers to the mass scale at which the axial vector current operator is renormalized. The power series in $\alpha_s(Q^2)$ in eq. (2.3) represents a perturbative calculation of the coefficient function of the axial vector currents [28, 29] and the $O(\Lambda^2/Q^2)$ terms are higher-twist corrections.

The first moment of g_1^p is special because to all orders in α_s there is no gluonic operator appearing on the right-hand side of eq. (2.3). An apparent candidate, a twist-2 spin-1 axial vector, is the anomalous current [30]

$$K_\mu = \epsilon_{\mu\alpha\lambda\sigma} \text{Tr} A^\alpha \left(F^{\lambda\sigma} - \frac{2}{3} A^\lambda A^\sigma \right). \quad (2.5)$$

However, this operator is not gauge invariant and cannot appear in an otherwise gauge invariant expression. The g_1^p sum rule is interesting because some combinations of Δu , Δd and Δs can be independently measured, or inferred, from data on nucleon and hyperon β -decays. The octet axial currents which mediate Gamow–Teller transitions between octet baryons are defined by

$$A_\mu^a \equiv \bar{\psi} \gamma_\mu \gamma_5 T^a \psi, \quad (2.6)$$

with $\text{tr} T^a T^b = \frac{1}{2} \delta^{ab}$. The nucleon matrix element of A_μ^3 is measured in neutron β -decay and determines $\Delta u - \Delta d$:

$$(\Delta u - \Delta d) s_\mu = 2 \langle p, s | A_\mu^3 | p, s \rangle = g_A s_\mu = (F + D) s_\mu. \quad (2.7)$$

g_A is the nucleon axial vector charge; $F + D$ is its parametrization in terms of SU(3) invariant matrix elements which is valid in the SU(3) symmetry limit. The nucleon matrix element of A_μ^8 is proportional to $\Delta u + \Delta d - 2\Delta s$, but the operator A_μ^8 does not appear in the β -decay hamiltonian. Its proton matrix element can be inferred from hyperon β -decay using SU(3) symmetry. In the SU(3)-limit,

$$(\Delta u + \Delta d - 2\Delta s)s_\mu = 2\sqrt{3} \langle p, s | A_\mu^8 | p, s \rangle = (3F - D)s_\mu. \quad (2.8)$$

We have dropped the label μ^2 on $\Delta\psi$ in eqs. (2.7) and (2.8) because these linear combinations are independent of renormalization point. The reason is that the currents A_μ^a would be conserved if quarks were massless, and that is enough to derive Ward identities which forbid them from being renormalized.

Comparing eqs. (2.7) and (2.8) with the sum rule, we see that the sum rule can be regarded as a measurement of the remaining combination of the $\Delta\psi$'s. It may be parametrized in terms of the flavor singlet operator

$$(\Delta u + \Delta d + \Delta s)s_\mu \equiv \Sigma(\mu^2)s_\mu = \left\langle p, s \left| \sum_{b=1}^3 \bar{\psi}_b \gamma_\mu \gamma_5 \psi_b \right|_{\mu^2} p, s \right\rangle, \quad (2.9)$$

whence

$$\int_0^1 dx g_1^p(x, Q^2) = \frac{1}{18}(3F + D + 2\Sigma(Q^2)) \left(1 - \frac{\alpha_s}{\pi} + O(\alpha_s^2) \right) + O\left(\frac{\Lambda^2}{Q^2}\right), \quad (2.10)$$

or equivalently in terms of $\Delta s(Q^2)$,

$$\int_0^1 dx g_1^p(x, Q^2) = \frac{1}{18}(9F - D + 6\Delta s(Q^2)) \left(1 - \frac{\alpha_s}{\pi} + O(\alpha_s^2) \right) + O\left(\frac{\Lambda^2}{Q^2}\right). \quad (2.11)$$

At this point it is important to note that Σ (and hence Δu , Δd and Δs individually) does depend on the renormalization point μ^2 . Unlike the flavor octet currents, the singlet current

$$A_\mu^0 = \sum_{b=1}^3 \bar{\psi}_b \gamma_\mu \gamma_5 \psi_b \quad (2.12)$$

is not conserved when quarks are massless; it has an anomalous divergence coming from the triangle graph [31]*

$$\partial^\mu A_\mu^0 = \frac{N_f \alpha_s}{2\pi} \text{Tr } F \tilde{F}, \quad (2.13)$$

* Throughout this paper we assume that the triangle diagram is defined so as to preserve the gauge invariance associated with the vector currents.

where N_f is the number of flavors. A_μ^0 is multiplicatively renormalized [32] – there is no gauge invariant axial vector gluon operator with which it can mix – and has an anomalous dimension which has been calculated to lowest non-trivial (two-loop) order [28].

In the early 1970's (before QCD) Ellis and Jaffe [33] made a prediction for $\int_0^1 g_1^p dx$ by suggesting that $\Delta s \cong 0$, i.e. that the polarized proton does not contain any polarized strange quarks. Using the values of F and D which we obtain in sect. 4

$$F = 0.47 \pm 0.04, \quad D = 0.81 \pm 0.03, \quad (2.14)$$

and taking $\alpha_s(Q^2) = 0.25 \pm 0.02$ [2] in the range of the EMC data, the up-to-date version of the Ellis–Jaffe sum rule is

$$\int_0^1 dx g_1^p(x, Q_{\text{EMC}}^2) = 0.175 \pm 0.018. \quad (2.15)$$

Note that the Ellis–Jaffe ansatz, $\Delta s \cong 0$, looks odd in light of QCD since Δs is scale-dependent and cannot vanish at both the confinement scale, $\mu_0 \leq 1$ GeV, and at the scale of the EMC experiment. Nevertheless, the idea that the strange quark content of the proton is small – a corollary of the OZI rule [18] – has become part of the folklore of hadronic physics and eq. (2.15) was taken as a benchmark by the experimenters.

The most recent result from EMC is [1]

$$\int_0^1 dx g_1^p(x) = 0.126 \pm 0.010 \pm 0.015. \quad (2.16)$$

If, for the purposes of discussion, we add the statistical and systematic errors in quadrature, and use eq. (2.14) for D and F , we find

$$\begin{aligned} \Delta u(Q_{\text{EMC}}^2) &= 0.78 \pm 0.08, & \Delta s(Q_{\text{EMC}}^2) &= -0.16 \pm 0.08 \\ \Delta d(Q_{\text{EMC}}^2) &= -0.50 \pm 0.08, \end{aligned} \quad (2.17)$$

or equivalently

$$\Sigma(Q_{\text{EMC}}^2) = 0.13 \pm 0.19. \quad (2.18)$$

These are the numbers which have aroused so much interest in the high-energy physics community. According to eq. (2.17), there must be a significant ($|\Delta s| \sim |\Delta u| \sim |\Delta d|$) component of polarized strange quarks in the proton. Eq. (2.18) has a superficially even more striking interpretation. As we shall see in sect. 6, the operator A_μ^0 measures the fraction of the spin of nucleons residing on the quark

spin. According to the EMC data that fraction is consistent with zero. The Ellis–Jaffe ansatz (at the EMC scale), $\Delta s(Q_{\text{EMC}}^2) \cong 0$, would have predicted

$$\Sigma(Q_{\text{EMC}}^2)_{\text{EJ}} = 3F - D = 0.60 \pm 0.12, \quad (2.19)$$

so the data are about two standard deviations away from the “folklore”.

Before taking this issue too seriously it is important to look for more mundane sources of the disagreement and for confirmation of the EMC result.

(i) The evaluation of the sum rule requires an extrapolation from the lowest measured x ($x_{\min} \cong 0.02$) to $x = 0$. Close and Roberts [2] have explored the sensitivity of $\Sigma(Q_{\text{EMC}}^2)$ to the assumed low- x behavior of g_1^p . They find it is not hard to invent a behavior singular enough to accommodate $\Delta s = 0$, however there seems to be little reason to expect such a singular behavior. It is evident from their analysis that it would be very useful to have an independent determination of Σ which does not require an extrapolation (see below).

(ii) The use of SU(3) symmetry for the axial vector charges is questionable. Unlike the vector charges, the axial charges are not protected from first-order corrections due to SU(3) violation [34]. SU(3) violations are known to be significant ($\approx 20\%$) for baryon magnetic moments [35]. However, SU(3) symmetry seems to work remarkably well for baryon axial charges. In fact, the major contribution to the uncertainty in F and D comes not from SU(3) violation but rather from discrepancies between different methods of determining axial charges in individual hyperon decays. We document these assertions in more detail in sect. 4.

(iii) As we have noted, Σ depends on the mass scale at which it is measured. It is possible that Σ is large (of order 0.6 as required by $\Delta s = 0$) at the confinement scale, $\mu_0 \lesssim 1$ GeV, but evolves away to a very small value $\Sigma \sim 0$ by $Q_{\text{EMC}}^2 \gtrsim 10$ GeV². This idea was presented and discussed in ref. [4]. In the perturbative domain, $Q^2 > 1\text{--}2$ GeV², Σ evolves slowly, its evolution near the confinement scale is unknown and could reconcile quark-model intuition with the deep inelastic measurements. In the absence of any idea of the non-perturbative evolution of Σ we must leave this caveat hanging.

(iv) Finally, shortly after the report of the EMC results, Kaplan and Manohar [5] showed that low-energy *elastic* neutrino–proton scattering could provide an independent measurement of another linear combination of Δu , Δd and Δs . Assuming SU(3) symmetry, they analyze existing data [6] and extract a value for $\Sigma(Q_{\text{EMC}}^2)$ which is consistent with the EMC result. Alternatively [36], one could accept the EMC result, in which case the $\nu p \rightarrow \nu p$ data confirm the reliability of SU(3) symmetry for the axial charges. Their result relies on some assumptions about the nucleon’s axial vector form factor which are subject to debate. A dedicated, high statistics, low-energy $\nu p \rightarrow \nu p$ scattering experiment could provide a much better measurement of $\Sigma(Q_{\text{EMC}}^2)$ [37]. More generally, the $\nu p \rightarrow \nu p$ experiment and analysis can be viewed as one step in an extensive low-energy program aimed at

obtaining a complete experimental picture of the flavor structure of the nucleon matrix elements of quark bilinears. We expand upon this idea in sect. 8.

3. The value of Σ expected in simple quark models

To appreciate the implications of the EMC result $\Sigma(Q_{\text{EMC}}^2) = 0.13 \pm 0.19$ for a quark model requires a careful formulation of the model in question. First consider the SU(6) symmetric non-relativistic quark model (NRQM)*. In the NRQM the proton is constructed from constituent quarks (U, D, S); they are not the current quarks (u, d, s) that appear in the QCD lagrangian. As a result, one has to rewrite the QCD currents in terms of the constituent quark fields. For example, the axial vector current in β -decay, $\bar{u}\gamma^\mu\gamma_5 d$ is rewritten as $g_A^{(8)}\bar{U}\gamma^\mu\gamma_5 D$ + higher derivative terms [38]. Here $g_A^{(8)}$ is an unknown renormalization that cannot yet be computed. The NRQM can then be used to calculate F and D in β -decay,

$$F = \frac{2}{3}g_A^{(8)}, \quad D = g_A^{(8)}. \quad (3.1)$$

The NRQM predicts $F/D = 2/3$, which is close to the experimental value $F/D = 0.58 \pm 0.05$. The experimental values for F and D determine the renormalization constant $g_A^{(8)}$ to be 0.75 ± 0.1 . There is a similar renormalization constant for the singlet current,

$$\bar{u}\gamma^\mu\gamma_5 u + \bar{d}\gamma^\mu\gamma_5 d + \bar{s}\gamma^\mu\gamma_5 s = g_A^{(0)}[\bar{U}\gamma^\mu\gamma_5 U + \bar{D}\gamma^\mu\gamma_5 D + \bar{S}\gamma^\mu\gamma_5 S] + \dots \quad (3.2)$$

$g_A^{(0)}$ and $g_A^{(8)}$ are independent; their difference is an indication of the importance of OZI-rule violations since only the singlet current can mix with gluons.

There are no constituent S quarks in the proton, so the NRQM predicts $\Sigma = g_A^{(0)}$. Thus the EMC experiment says that $g_A^{(0)} = 0.13 \pm 0.19$. Since $g_A^{(8)} = 0.75 \pm 0.1$, there appears to be large OZI violation in the axial vector sector. There are no s (current) quarks in the proton if $g_A^{(0)} = g_A^{(8)}$, i.e. if $\Sigma = 0.75 \pm 0.1$. This is the Ellis–Jaffe value ($\Sigma_{\text{EJ}} = 0.60 \pm 0.12$) derived in sect. 2. [The numbers do not agree exactly because here F/D has been constrained to be $2/3$ in contrast to the Ellis–Jaffe analysis.]

There has been some discussion about contradictions between the EMC and the NRQM [39]. In these discussions the constituent quarks were not distinguished from the QCD quarks. In this case, one gets $\Sigma = 1$. However this model also predicts [40] $F = 2/3$ and $D = 1$ which is not in agreement with hyperon β -decay. The model has more serious problems than this; e.g. it predicts that the structure

* The next three paragraphs paraphrase the discussion of ref. [5].

function F_1 is proportional to $\delta(x - \frac{1}{3})$. In the discussion of axial vector currents it is important to distinguish between current and constituent quarks in the NRQM. We will not discuss further the version of the NRQM in which the constituent and current quarks are identified.

Bag models provide an alternative description of hadron phenomenology [41]. Bag quarks are identified with QCD quarks and the effects of confinement and chiral symmetry violation are lumped into boundary conditions. Bag quarks have (small) current-quark masses, so relativistic effects cannot be ignored [41–43]. In a relativistic quark model the orbital contribution to the angular momentum is important even in the lowest partial wave. In particular, the lower (Dirac) components are spinning (on average) in a direction opposite to the upper components. For example, in a state with $j = \frac{1}{2}$ (and even parity) the upper components of the Dirac spinor have $l = 0$ but the lower components have $l = 1$ coupled to $s = \frac{1}{2}$ to give total $j = \frac{1}{2}$. In the state with $m_j = \frac{1}{2}$ the expectation value of s_z in the lower component of the Dirac wave function is $-\frac{1}{6}$. If we describe the bag quarks bound in a proton by the Dirac wave function $\begin{pmatrix} f \\ \sigma \cdot \hat{r} g \end{pmatrix}$ with $\int(f^2 + g^2) = 1$, then in the SU(3) limit [44]

$$F = \frac{2}{3} \int (f^2 - \frac{1}{3}g^2), \quad D = \int (f^2 - \frac{1}{3}g^2) \quad (3.3)$$

where $\langle \sigma \cdot \hat{r} \sigma^3 \sigma \cdot \hat{r} \rangle = -\sigma^3/3$ is the crucial calculation. One may regard $\int(f^2 - \frac{1}{3}g^2)$ as a relativistic “depolarizing” factor parametrizing the extent to which kinematic relativistic $l \cdot s$ coupling decouples the quark spin from its total angular momentum. The data on g_A require $\int(f^2 - \frac{1}{3}g^2) \approx 0.75$ and relativistic quark models like the bag model have no trouble generating this. In fact, the bag model with massless quarks overshoots, $\int(f^2 - \frac{1}{3}g^2)_{\text{bag}} = 0.65$. The exact size of this effect has been used to estimate effective quark masses in relativistic quark models [45]. Note that this relativistic effect does not affect F/D (at least in the SU(3) limit) so the small deviation of F/D from $2/3$ is a problem for these models.

So, in relativistic quark models the deficit of Σ from unity is merely the fraction of the nucleon’s spin carried by the quark’s orbital angular momentum [43]. This interpretation of the Ellis–Jaffe sum rule was first suggested by Sehgal in 1974 [46]. Of course, the proton is likely to be much more complicated than either the bag model or NRQM suggest. However, it should be recognized that both of these minimal quark models (consistent with octet baryon β -decays) already suggested that 25–40% of the proton’s angular momentum is not due to the quark’s spin.

The lesson of this section is a simple one: unity is not a “naive” expectation for Σ , it is an unreasonable one. A more reasonable “zeroth”-order expectation, based only on SU(3) symmetry and the absence of polarized s-quarks in the proton, is $\Sigma = 3F - D = 0.60 \pm 0.12$. In a “naive” quark model about 60% of the proton’s

spin is carried by quark spin, the other 40% is carried by quark orbital angular momentum. The NRQM quark model can accommodate the EMC results if one includes OZI-rule violation, i.e. allows $g_A^{(8)} \neq g_A^{(0)}$. The bag model can also, if one includes quark annihilation effects.

4. F and D matrix elements

The F and D invariant matrix elements of the axial current are extracted from the measured values of the hyperon semileptonic decay rates and/or polarization asymmetries. The corrections to the standard F and D parametrization come from SU(3) breaking, from q^2 dependence in the form factors, and from second-class currents. These last two corrections are of the same order of magnitude as SU(3) breaking. We will calculate the values of F and D without including any SU(3) breaking effects; thus we also leave out q^2 dependence in the form factors and second-class currents. We will also leave out weak magnetism, since its effect on the total decay rate is unimportant [47]. We take asymmetry and lifetime data from the latest Particle Data Group publication [48]. Unfortunately, the values of g_A obtained from the lifetime and polarization asymmetry disagree. This discrepancy between the two determinations of g_A is presumably due to systematic errors in the experiments. We deal with this as follows: if the lifetime implies $g_A = \alpha \pm \Delta\alpha$ and the asymmetry gives $g_A = \beta \pm \Delta\beta$, we take $(g_A)_{\max} = \max(\alpha + \Delta\alpha, \beta + \Delta\beta)$, $(g_A)_{\min} = \min(\alpha - \Delta\alpha, \beta - \Delta\beta)$, and use $g_A = \frac{1}{2}[(g_A)_{\max} + (g_A)_{\min}]$, $\Delta g_A = \frac{1}{2}[(g_A)_{\max} - (g_A)_{\min}]$. Typically this increases the errors on g_A by about a factor of three. For the decays (such as $\Lambda \rightarrow p\mu\bar{\nu}_\mu$) where there is only a lifetime determination of g_A , we have multiplied the errors by three so that these decays do not contribute unduly to the fit. Performing a weighted least squares fit to the modified data yields $D = 0.81 \pm 0.03$, $F = 0.47 \pm 0.04$. The F and D values allowed by the various electron decay modes are shown in fig. 1. The dashed curve is the line $3F - D = 0$, for which one can have $\Sigma = 0$ and $\langle p|\bar{s}\gamma^\mu\gamma_5 s|p\rangle = 0$ simultaneously. The best fit point is approximately two standard deviations away from this line. There is no evidence in the fit for any large SU(3) breaking effects in the data ($\chi^2 = 4$ for nine degrees of freedom, suggesting our procedure for estimating errors is overly conservative). We would typically expect SU(3) breaking effects to be of order 25%; they appear to be smaller for the F and D matrix elements of the axial charge. Often much smaller error bars are quoted on F and D (see, for example, ref. [1]). Even if all hyperon β -decay measurements were consistent within very small errors with a single choice of F and D , we would argue that the error assigned to $\Delta u + \Delta d - 2\Delta s$ should be enlarged, since we have no reliable estimate of the accuracy of the SU(3) rotation to the hypercharge ($a = 8$) direction and no reason to expect SU(3) violation to be as small as the often quoted values of F and D would suggest.

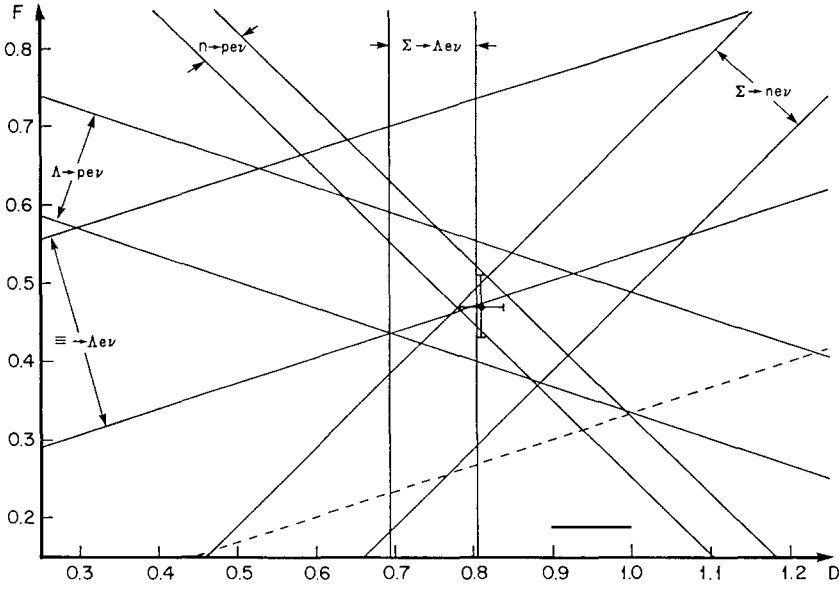


Fig. 1. Allowed regions of F and D for various hyperon β decays. The point marked is the best fit value. The dashed line $3F - D = 0$ marks the points where one can have no strange quarks in the proton and $\Sigma = 0$ simultaneously.

5. g_1 in QCD

There are three parts to this section. First we discuss the OPE analysis of $\int dx g_1^p(x, Q^2)$ in QCD. We review the standard analysis and then turn to the work of Altarelli and Ross [12], Altarelli and Stirling [49] and Carlitz, Collins and Mueller [13, 50], which we criticise in the framework of the OPE. Second we study the Ward identity obeyed by the flavor singlet axial vector current, A_μ^0 . This reveals an ambiguity we believe afflicts an attempt to define a “gluonic” contribution to $\Sigma(\mu^2)$. It also makes clear that an analysis which is not sufficiently careful about the U(1)-problem in QCD could erroneously arrive at a conclusion very like that of refs. [12, 13]. Finally, we explicitly compute the contribution of $\int dx g_1^p(x, Q^2)$ from “fairly heavy” quarks, $Q^2 > m^2 \gg \Lambda_{\text{QCD}}^2$, to illustrate the pitfalls of attempting to extract a “gluonic” contribution to the sum rule.

5.1. $g_1(x, Q^2)$ AND THE OPE

The OPE analysis of $\int dx g_1^p(x, Q^2)$ in QCD is in one sense remarkably simple: this sum rule projects out a unique operator, $\bar{\psi} \gamma_\mu \gamma_5 \mathcal{Q}^2 \psi$, where $\mathcal{Q}$ is the quark charge matrix; there is no gauge invariant gluonic operator with this dimension and spin and therefore no operator mixing problem. The coefficient function $\bar{\psi} \gamma_\mu \gamma_5 \mathcal{Q}^2 \psi$ has been calculated to order α_s [29]. The non-singlet part of the operator has no

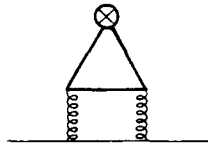


Fig. 2. The two-loop graph which produces an anomalous dimension for the singlet axial vector current.

anomalous dimension; the singlet part has an anomalous dimension which has been calculated to lowest non-trivial order [28]. Finally, the nucleon matrix elements of $\bar{\psi}\gamma_\mu\gamma_5\mathcal{Q}^2\psi$ cannot be calculated perturbatively, at present the best we can do is study it in models such as those discussed in sect. 3.

The non-singlet contributions to the sum rule are not controversial so we drop them and concentrate on

$$s_\mu \Sigma(Q^2) = \langle p, s | A_\mu^0 | Q^2 | p, s \rangle. \quad (5.1)$$

The Q^2 -dependence of $\Sigma(Q^2)$ is slight in the perturbative domain. Because A_μ^0 is multiplicatively renormalized, $\Sigma(Q^2)$ evolves homogeneously with Q^2 (cf. refs. [4, 12, 13]). Because its anomalous dimension begins at two-loop order (fig. 2) it evolves more slowly than other twist-two operators with one-loop anomalous dimensions;

$$\Sigma(Q^2) = \left\{ 1 - \frac{1}{\pi} \frac{6N_f}{33 - 2N_f} (\alpha_s(\mu^2) - \alpha_s(Q^2)) \right\} \Sigma(\mu^2). \quad (5.2)$$

Eq. (5.2) is sufficient to run $\Sigma(Q^2)$ down from the large Q^2 where it is measured by EMC to as low a μ^2 as one is willing to trust next-to-leading order renormalization group improved QCD perturbation theory. The slow decrease of $\Sigma(Q^2)$ reflects the slow transfer of quark spin to quark orbital angular momentum via gluon intermediate states made possible by the anomaly. This is described clearly and accurately, albeit in the service of another line of argument, in ref. [13].

An important issue, first raised by Efremov and Teryaev [11] and then studied in detail by Altarelli and Ross [12] and Carlitz et al. [13] is whether perturbative QCD can be used to make any further statements about $\Sigma(Q^2)$. Ref. [11] contains mistakes which were subsequently corrected in refs. [12, 13]. Refs. [12, 13] differ in the details, but both propose that the small value of $\Sigma(Q^2)$ can be understood as a cancellation of an “intrinsic” quark contribution, $\tilde{\Sigma}(Q^2)$, by a “gluonic” contribution $\Delta\Gamma(Q^2)$, given by

$$\Delta\Gamma(Q^2) = -\frac{\alpha_s(Q^2)}{2\pi} N_f \Delta g(Q^2), \quad (5.3)$$

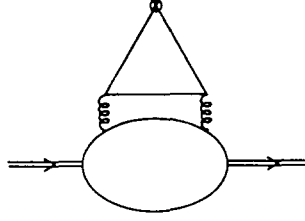


Fig. 3. Coupling of A_μ^0 to the proton through the triangle diagram.

where Δg is an integrated gluon distribution^{*},

$$\Delta g(Q^2) = \int_0^1 dx [g_\uparrow(x, Q^2) - g_\downarrow(x, Q^2)]. \quad (5.4)$$

Carlitz et al. [13] especially (although, see also ref. [24]) stress that $\Delta\Gamma(Q^2)$ is measurable as a two-quark jet contribution to $g_1^p(x, Q^2)$. The idea then is that

$$\Sigma(Q^2) = \tilde{\Sigma}(Q^2) - \frac{\alpha_s(Q^2)}{2\pi} N_f \Delta g(Q^2) \quad (5.5)$$

with $\tilde{\Sigma}(Q^2)$ being quark-model like (i.e. ≈ 0.60) and approximately cancelled by the large gluon contribution, $\Delta\Gamma(Q^2)$.

We agree with the authors of refs. [13, 24] that two-quark jet production in e^+p^+ scattering can be *reliably* calculated in an appropriate kinematic region and that the resulting contribution to $g_1^p(x, Q^2)$ is given by eq. (5.3). We disagree on two other points: first, this is not the only “gluonic” contribution to $\Sigma(Q^2)$. Any attempt to isolate a gluonic contribution to $\Sigma(Q^2)$ is beset by ambiguities (which are illustrated by the analysis of U(1) Ward identities given in the following subsections). And second, separating out the two-quark jet piece, $\Delta\Gamma(Q^2)$, leaves behind a piece $\tilde{\Sigma}(Q^2)$ which cannot be considered the “intrinsic” quark contribution. In general it contains gluonic contributions and it is large for “fairly heavy” quarks which should not appear at all in the “intrinsic” description of the nucleon. Once $\tilde{\Sigma}(Q^2)$ has lost its significance as the “intrinsic” quark contribution, eq. (5.5) is left without content.

To some extent, the argument of refs. [12, 13] rests on an analysis of those contributions to the matrix element of A_μ^0 which involve the triangle diagram (see fig. 3). They claim, at least in the case of massless quarks, first, that the triangle diagram contributes $\Delta\Gamma(Q^2)$ to the sum rule, and second that this is a reliable calculation in QCD.

^{*} Our notation differs from refs. [12, 13]. The quantities we call Σ , $\tilde{\Sigma}$ and Δg are called $\Delta q'$, Δq and Δg in ref. [13] and M_1^s , $\Delta\Sigma$ and Δg in ref. [12].

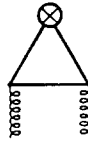


Fig. 4. A graph contributing to the gluon matrix element of a quark operator.

Their argument is based on a calculation of the gluon matrix element of the operator A_μ^0 (fig. 4), or equivalently of the box diagram which gives

$$\langle g, s | A_\mu^0 | g, s \rangle = \begin{cases} -\alpha_s/2\pi & p^2 \gg m^2, \\ 0 & p^2 \ll m^2, \end{cases} \quad (5.6)$$

where p^2 and m^2 are the gluon invariant mass-squared and the quark mass-squared, respectively. Carlitz et al. observe that the $-\alpha_s/2\pi$ contribution comes from $k_\perp \sim \sqrt{Q^2}$ in the Compton process and is present whether or not p^2 is large compared to m^2 . This large $k_\perp$ process would be observed as “ γ ” $p \rightarrow 2$ jets and measures a (polarized) gluon distribution Δg . When $m^2 \gg p^2$, infrared contributions give rise to a $+\alpha_s/2\pi$ contribution which exactly cancels this “hard” contribution. However, as they explain, this contribution is part of the “ γ ” $p \rightarrow 1$ jet cross section, measures a quark distribution, and belongs in $\Sigma(Q^2) - \Delta\Gamma(Q^2) = \tilde{\Sigma}(Q^2)$ which they view as the “intrinsic” quark contribution to $\Sigma(Q^2)$. The reader is referred to ref. [13] for a more complete summary of this argument.

The authors of refs. [12, 13] suggest that the triangle diagram provides a *local* probe of the gluon distribution in the target. If this were true, $\Delta\Gamma$ would be protected from infrared problems and the calculation would be reliable in the usual sense. However, we believe there are strong arguments that the triangle is not local in the sense required. It is therefore not necessarily protected from infrared effects, in particular from the non-perturbative effects which give the η' a mass^{*}.

The general form of the triangle graph in which the axial current and gluons carry momenta q , k_1 and k_2 ($q = k_1 + k_2$) is given, for example, by Adler [52]. It is clearly non-local (i.e. not a polynomial in k_1 and k_2). The arguments of refs. [12, 13] do not require the entire triangle to be local, only the “anomalous part” which has no Fock-space interpretation and gives rise to the anomalous divergence. However, the essence of the anomaly is the proof (see ref. [52]) that there is no gauge invariant local counterterm for the anomaly. Thus we know for certain that the “anomalous part” of the triangle which gives rise to the $-\alpha_s/2\pi$ cannot

^{*}In an earlier version of this paper we suggested that the identification of $\Delta\Gamma$ as a “gluonic” contribution would break down in higher orders of perturbation theory and proposed graphs which might be responsible. Subsequently, explicit calculations have shown this not to be the case [50].

be described by a gauge invariant local operator. If, however, one considers only forward matrix elements ($k_1 = k_2 = k$), the triangle collapses to the simple form (for massless quarks)

$$R_{\mu\alpha\beta}(k, k) = + \frac{\alpha_s}{2\pi} \epsilon_{\mu\lambda\alpha\beta} k^\lambda, \quad (5.7)$$

which appears local (α and β are gluon Lorentz indices). The apparent locality of the forward matrix element means nothing. It is trivial to construct even simpler examples of non-local operators with local forward matrix elements.

The forward matrix element, $R_{\mu\alpha\beta}(k, k)$, coincides with the forward two-gluon matrix element of the local current,

$$k_\mu = \frac{\alpha_s}{2\pi} K_\mu = \frac{\alpha_s}{2\pi} \epsilon_{\mu\nu\alpha\beta} \text{Tr } A^\nu \left[F^{\alpha\beta} - \frac{2}{3} A^\alpha A^\beta \right], \quad (5.8)$$

and the authors of refs. [12, 13] advocate using k_μ as a local interpolating field for the “gluonic contribution” to $\Sigma(\mu^2)$. There are two problems with this replacement: First, the triangle is not local whereas k_μ is. Thus the triangle is likely to be sensitive to the infrared domain in a way in which k_μ is not. Second, neither k_μ nor its forward matrix element is gauge invariant in QCD. In an abelian gauge theory the gauge dependence of k_μ is a total derivative which has vanishing forward matrix element. In QCD

$$k_\mu \rightarrow k_\mu + \frac{\alpha_s}{2\pi} \epsilon_{\mu\nu\alpha\beta} \text{Tr } U^\dagger \partial^\nu U U^\dagger \partial^\alpha U U^\dagger \partial^\beta U + \text{total derivatives}, \quad (5.9)$$

and the inhomogeneous term, though still a total derivative, does not vanish for “large” gauge transformations (i.e. those which change instanton number).

Since the calculation clearly gives rise to the triangle graph rather than k_μ and since the locality and gauge invariance properties of the two are likely to be different, we believe the replacement of the former by the latter to be unjustified. Henceforth, we will stick with the gauge invariant, non-local triangle diagram itself.

5.2. $U(1)_A$ WARD IDENTITIES

The anomalous conservation law, eq. (2.13), can be used to relate Σ to the near-forward matrix element of the gluon operator $(N_f \alpha_s / 2\pi) \text{Tr } F\tilde{F}$. The analysis is a standard one, gauge invariant at every step. For simplicity we consider massless quarks. The case $m \ll \Lambda$ yields similar results. Define the form factors for the coupling of A_μ^0 to the nucleon

$$\langle N(p's) | A_\mu^0 | N(ps) \rangle = \Sigma(t) s_\mu + l \cdot s l_\mu h(t), \quad (5.10)$$

where $l_\mu = p'_\mu - p_\mu$, $t \equiv l^2$ and $p \cdot s = p' \cdot s = 0$. We have used an abbreviated notation here. $\Sigma = \Sigma(Q^2, t)$ depends on both the renormalization scale, Q^2 , and the invariant four momentum transfer, t . Since Q^2 plays no role in the discussion of this section, we suppress it, $\Sigma(Q^2, t) \rightarrow \Sigma(t)$. Thus $\Sigma(0)$ in this section is identical to $\Sigma(Q^2)$ in the previous discussion. $\Sigma(t)$ and $h(t)$ are the only form factors allowed by the usual symmetries of QCD. Likewise, parametrize the coupling of $F\tilde{F}$ to the nucleon by

$$\left\langle N(p's) \left| \frac{N_f \alpha_s}{2\pi} \text{Tr} F\tilde{F} \right| N(ps) \right\rangle \equiv i l \cdot s \kappa(t). \quad (5.11)$$

From eq. (2.13) we obtain an anomalous Ward identity

$$\Sigma(t) + th(t) = \kappa(t). \quad (5.12)$$

It will be useful to compare with the analogous flavor non-singlet Ward identity

$$\Delta q^a(t) + th^a(t) = 0 \quad (5.13)$$

for the operator A_μ^a , $a = 1, 2, \dots, 8$.

Now take $t \rightarrow 0$. $h(t)$ has no pole at $t = 0$ because the η' is massive even when all quarks are massless. Thus

$$\Sigma(0) = \kappa(0). \quad (5.14)$$

This is to be compared with

$$\Delta q^a(0) = -f^a(0), \quad (5.15)$$

where $f^a(0)$ is the residue of the Goldstone boson pole in $h^a(t)$ at $t = 0$, i.e. $h^a(t) = f^a(0)/t + \text{non-singular terms}$.

Eq. (5.14) makes it clear that there is no distinction between Σ and the nucleon matrix element of the gluonic operator $F\tilde{F}$. Another example of this ambiguity is well known: Consider the scale current, d_μ , with divergence

$$\partial^\mu d_\mu = T_\alpha^\alpha = \frac{\beta(g)}{g} \text{Tr} F^2. \quad (5.16)$$

Since the forward matrix element of the stress tensor, $T^{\mu\nu}$, is normalized by

$$\langle p | T^{\mu\nu} | p \rangle = 2p^\mu p^\nu, \quad (5.17)$$

we find a Ward identity

$$2M^2 = \left\langle p \left| \frac{\beta(g)}{g} \text{Tr } F^2 \right| p \right\rangle. \quad (5.18)$$

Eqs. (5.14) and (5.18) have a similar interpretation: if it were not for gluon-mediated interactions, the nucleon would consist of three free, massless quarks. The ground state mass and axial charge of a “nucleon” composed of free massless quarks vanishes, so it is not surprising that either can be expressed (by means of the equations of motion) in terms of gluon operators.

Suppose one attempted to evaluate $\kappa(0)$ in the parton model. A brief calculation (inserting $(N_f \alpha_s / 2\pi) \text{Tr } F\tilde{F}$ in gluon Fock states) yields

$$\Sigma_{\text{parton}}(0) = \kappa_{\text{parton}}(0) = -\frac{\alpha_s}{2\pi} N_f \Delta g,$$

where Δg is the gluon helicity asymmetry defined in eq. (5.4). Comparing this result with the separation proposed in refs. [12, 13] (see eq. (5.5)), two choices appear: (1) $\tilde{\Sigma} = 0$, or (2) the parton model calculation of $\kappa(0)$ is incomplete. A possible non-partonic contribution to Σ has recently been discussed by Forte [53].

Comparing eqs. (5.14) and (5.15) it is clear how one might *erroneously* arrive at eq. (5.5), the central result of refs. [12, 13]. Suppose one missed the effects that give the η' a mass. This would replace eq. (5.14) by

$$\Sigma(0) = -f(0) + \kappa(0), \quad (5.14a)$$

where $f(0)$ is the residue of the η' -pole in $h(t)$. Next suppose one evaluated $\kappa(0)$ in the parton model as given by eq. (5.18). The result would be

$$\Sigma = -f - \frac{\alpha_s}{2\pi} N_f \Delta g. \quad (5.14b)$$

Comparing eq. (5.15) one would identify $-f$ as the “intrinsic” quark contribution (the non-singlet axial charges only get these contributions) and the remainder as the intrinsic gluonic contribution. Whether the resemblance of eq. (5.14b) to eq. (5.5) is more than coincidental is hard to say since we are not able to trace the non-perturbative effects such as instantons (which give the η' a mass) or the deviations from κ_{parton} through the parton model calculation.

Finally we turn to a technical question: how can $\kappa(0)$ enter the *forward* matrix element of A_μ^0 when the matrix element of $F\tilde{F}$ explicitly vanishes in the forward direction? The answer lies in the triangle diagram. For massless quarks and on-shell gluons, the off-forward matrix element of the triangle diagram (see fig. 3) coincides with the matrix element of $-i(l^\mu/l^2)(\alpha_s/2\pi)\text{Tr } F\tilde{F}$ [54]. This result is

regularization-independent. In QCD, the pole at $l^2 = 0$ is unphysical and is cancelled by non-triangle contributions to the matrix element of A_μ^0 . With the aid of this we can decompose the matrix element of A_μ^0 into the contribution from the triangle and the rest, using eq. (5.10),

$$\Sigma(t)s^\mu + l \cdot sl^\mu h(t) = \frac{l \cdot sl^\mu}{l^2} \kappa(t) + \left(s^\mu - \frac{l \cdot sl^\mu}{l^2} \right) \lambda(t), \quad (5.19)$$

where the first term on the right comes from the triangle and the second is the rest. The cancellation of the anomaly pole, as well as the Ward identity, eq. (5.14), requires $\lambda(0) = \kappa(0)$. Note that the triangle contribution is singular as $l^\Lambda \rightarrow 0$: the answer depends on the direction in which the limit is taken. [Of course, $\Sigma(0)$ is finite and independent of the direction of the limit.] Apparently, the (forward) calculation of refs. [12, 13] is equivalent to taking the limit so that $l \cdot sl^\mu / l^2 \rightarrow s^\mu$.

To summarize: a study of the $U(1)_A$ Ward identity in QCD for $m = 0$ reveals that it is impossible in principle to separate a gluonic and an “intrinsic quark” contribution to Σ . It also suggests how, ignoring certain non-perturbative effects, one could erroneously arrive at the separation proposed in refs. [12, 13].

5.3. THE CONTRIBUTION TO $fg_1^p(x, Q^2)$ FROM A HEAVY QUARK

In this subsection we study the contribution to $\int dx g_1^p(x, Q^2)$ from a quark of mass considerably larger than Λ_{QCD} but still small enough to be produced in a (hypothetical) deep inelastic scattering experiment: $Q^2 \gg m^2 \gg \Lambda_{\text{QCD}}^2$. For definiteness one could consider the charm-quark (from which we borrow our notation).

First we review the picture of refs. [12, 13]:

$$\Delta c = \widetilde{\Delta c} - \frac{\alpha_s}{2\pi} \Delta g. \quad (5.20)$$

The second term, observable in polarized two-jet production, is independent of m (since $Q^2 \gg m^2$) and must be large in order to compensate $\widetilde{\Sigma} \approx 0.6$ and produce $\Sigma \approx 0$. $\widetilde{\Delta c}$ is the contribution of “intrinsic” heavy quarks, but there are none in the quark model description of the nucleon so we expect $\widetilde{\Delta c} \approx 0$.

For heavy quarks ($M \gg \Lambda_{\text{QCD}}$) it is possible to estimate the size of Δc (not $\widetilde{\Delta c}$) using effective field theories. The relevant calculation was done in appendix B of ref. [4]. The result can be summarized here: the heavy quark in the operator $O^\mu = \bar{c} \gamma^\mu \gamma_5 c$, can be integrated out to give

$$\begin{aligned} O^\mu &= \frac{\alpha_s(m)}{6\pi^2 m^2} \epsilon_{\mu\rho\tau\sigma} \text{Tr} [F^{\alpha\rho} \partial_\alpha F^{\tau\sigma} + 2F^{\tau\sigma} \partial_\alpha F^{\alpha\rho}] \\ &+ O\left(\frac{\alpha_s(m)}{m^2}, \frac{\alpha_s^2(m)}{m^2}\right). \end{aligned} \quad (5.21)$$

Here $\alpha_s(m)$ is the strong interaction coupling constant at the heavy quark scale, and $F^{\mu\nu}$ is the gluon field strength tensor. Thus the proton matrix element of O^μ is obtained by taking the proton matrix element of the right-hand side of eq. (5.21), with the result

$$s^\mu \Delta c = \langle p, s | O^\mu | p, s \rangle = O \left(\frac{\Lambda_{\text{QCD}}^3}{m_f^2} \right), \quad (5.22)$$

since proton matrix elements of the gluon field strength operators are of order Λ_{QCD}^3 . Thus we find

$$\Delta c = O \left(\frac{\Lambda_{\text{QCD}}^2}{m_f^2} \right) = \widetilde{\Delta c} - \frac{\alpha_s}{2\pi} \Delta g, \quad (5.23)$$

in contradiction to the picture proposed in refs. [12,13]: since $\Delta c \approx 0$, it is not possible to have $(\alpha_s/2\pi)\Delta g$ large unless $\widetilde{\Delta c}$ is comparable.

Finally, it is instructive to use the methods of ref. [13] to see how $\Delta c \approx 0$ comes about. The calculation of the box diagram reveals two distinct contributions to $g_1^p(x, Q^2)$ from a massive quark. One, peaked at $k_\perp \approx \sqrt{Q^2}$ contributing to two-jet production, is put in the gluonic contribution. The other, peaked at $k_\perp \approx m$, is lumped into $\widetilde{\Delta c}$. The two contributions explicitly cancel to the order displayed in by eq. (5.23).

6. Angular momentum, spin sum rules, and the anomalous current K_μ

In this section we review the operator description of angular momentum in a theory of quarks and gluons. We derive the sum rule for the nucleon's spin and relate the terms that appear there to local operators and to quark and gluon distribution functions in an infinite momentum frame. Some of the issues we must confront are:

- (i) What is the operator description of the spin and orbital angular momentum of quarks and gluons?
- (ii) How are the matrix elements of the angular momentum density tensor, $M^{\mu\nu\lambda}$, normalized?
- (iii) What is the relation between the gluon spin and the anomalous current

$$K_\sigma = \epsilon_{\sigma\alpha\beta\gamma} \text{Tr} A^\alpha \left(F^{\beta\gamma} - \frac{2}{3} A^\beta A^\gamma \right). \quad (6.1)$$

- (iv) Is there a sum rule for the nucleon's angular momentum in an infinite momentum frame and does it get a contribution from the orbital angular momentum of quarks and gluons?

We apologize to readers to whom these issues appear elementary and/or well known. We have found some confusion surrounding each of them in recent work on g_1 .

6.1. THE TENSOR $M^{\mu\nu\lambda}$ AND A GENERAL ANGULAR MOMENTUM SUM RULE

The current associated with translations is the stress tensor, $T_{\mu\nu}$, which is conserved, $\partial_\mu T^{\mu\nu} = 0$, symmetric, $T_{\mu\nu} = T_{\nu\mu}$, and gauge invariant. The current associated with Lorentz transformations is a rank-3 tensor constructed entirely from $T_{\mu\nu}$ [55],

$$M^{\mu\nu\lambda} \equiv x^\nu T^{\mu\lambda} - x^\lambda T^{\mu\nu}. \quad (6.2)$$

$M^{\mu\nu\lambda}$ is conserved, $\partial_\mu M^{\mu\nu\lambda} = 0$, because $T^{\mu\nu}$ is symmetric and conserved. $M^{\mu\nu\lambda}$ is gauge invariant, and has no totally antisymmetric part,

$$\epsilon_{\alpha\mu\nu\lambda} M^{\mu\nu\lambda} = 0, \quad (6.3)$$

or equivalently

$$M^{\mu\nu\lambda} + M^{\lambda\mu\nu} + M^{\nu\lambda\mu} = 0. \quad (6.4)$$

It is a textbook exercise to verify that the conserved charges,

$$J^{\nu\lambda} \equiv \int d^3x M^{0\nu\lambda}, \quad (6.5)$$

$$P^\mu \equiv \int d^3x T^{0\mu}, \quad (6.6)$$

obey the algebra of the Poincaré group. There are no quantum anomalies in this algebra (at least for ordinary gauge theories in flat four-dimensional space-time).

We wish to study the single nucleon matrix element of $M^{\mu\nu\lambda}$ in hopes of relating it to the nucleon's spin. Because of the explicit factors of x^μ in the definition of $M^{\mu\nu\lambda}$, its matrix elements are rather more singular than operators which may be more familiar, and more care must be taken with them. To warm up, we consider the properties of $T_{\mu\nu}$. Let $|p, s\rangle$ be a plane wave nucleon state with momentum p_μ and spin s_μ , normalized to $\langle p, s | p', s' \rangle = 2E(2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}')$. Define a form factor for an insertion of $T_{\mu\nu}$ with momentum k on a single nucleon line:

$$\mathcal{T}_{\mu\nu}(p, k, s) \equiv \int d^4x \langle p', s | T_{\mu\nu}(x) | p, s \rangle e^{ik \cdot x}. \quad (6.7)$$

For reasons which will become clear shortly, we do not consider the most general case but instead constrain $s' = s$. We translate $T_{\mu\nu}$ to $x = 0$ and perform the

x -integration,

$$\mathcal{T}_{\mu\nu}(p, k, s) = (2\pi)^4 \delta^4(p' + k - p) \langle p', s | T_{\mu\nu}(0) | p, s \rangle. \quad (6.8)$$

$T_{\mu\nu}$ is conserved and symmetric and is a tensor (not pseudotensor). The matrix element can at most be linear in the spin s_μ . Although many tensors constructed from p^μ , k^μ , s^μ , $g^{\mu\nu}$ and $\epsilon^{\mu\lambda\alpha\beta}$ can appear on the right-hand side of eq. (6.8), we are only going to need those which are independent of k^μ or linear in k^μ :

$$\langle p', s | T_{\mu\nu}(0) | p, s \rangle = A_0(k^2) P_\mu P_\nu + iA_1(k^2) (\epsilon_{\mu\alpha\beta\sigma} P_\nu + \epsilon_{\nu\alpha\beta\sigma} P_\mu) k^\alpha P^\beta s^\sigma + O(k^2), \quad (6.9)$$

where $P^\mu = \frac{1}{2}(p + p')^\mu$. Both of these terms are conserved ($k^\mu P_\mu = 0$), symmetric, and have the correct parity. An example of a form factor which is ignored is $(k^2 g_{\mu\nu} - k_\mu k_\nu) A_2(k^2)$. It makes a vanishingly small contribution near $k^2 = 0$ since there are no massless particles in QCD. We now take $k^\mu \rightarrow 0$,

$$\mathcal{T}_{\mu\nu}(p, 0) = (2\pi)^4 \delta^4(0) A_0(0) p_\mu p_\nu. \quad (6.10)$$

Next we use the fact that T^{00} is related to the hamiltonian, $H \equiv \int d^3x T^{00}(x)$, to derive

$$\mathcal{T}_{00}(p, 0) = 2(2\pi)^4 \delta^4(0) E^2 \quad (6.11)$$

directly from the definition (6.7). Comparing eqs. (6.10) and (6.11) we find $A_0(0) = 2$, so

$$\langle p, s | T_{\mu\nu}(0) | p, s \rangle = 2p_\mu p_\nu. \quad (6.12)$$

This normalization condition is well known and is the basis of the ‘‘momentum’’ sum rule of deep inelastic phenomenology [19]. The reader may wonder why we have taken such care with $T_{\mu\nu}$ since we could have worked entirely in coordinate space and with forward matrix elements and obtained the same result much more quickly. The answer is that exactly this amount of care is necessary to avoid mistakes* with $M^{\mu\nu\lambda}$.

* Ref. [11] reaches erroneous conclusions by an oversimple analysis of $M^{\mu\nu\lambda}$. Here, for example is a (erroneous) proof that the proton has no spin: Consider $\langle p, s | M^{\mu\nu\lambda} | p, s \rangle$. Use eq. (6.2) to define $M^{\mu\nu\lambda}$ and eq. (6.12) for the matrix element $\langle p, s | T^{\mu\nu} | p, s \rangle$, to obtain $\langle p, s | M^{\mu\nu\lambda}(x) | p, s \rangle = 2(x^\nu p^\mu p^\lambda - x^\lambda p^\mu p^\nu)$. The angular momentum J^k is the space integral of $\frac{1}{2}\epsilon^{kij} M^{0ij}$. But in the rest frame, $\langle p, s | M^{0ij}(x) | p, s \rangle \equiv 0$! What is wrong with this argument is that the $k \rightarrow 0$ limit (always understood since the true states are wave packets) does not commute with the x integral.

Consider the momentum space form factor for an insertion of $M^{\mu\nu\lambda}$ on a nucleon line:

$$\mathcal{M}^{\mu\nu\lambda}(p, k, s) \equiv \int d^4x e^{ik \cdot x} \langle p', s | M^{\mu\nu\lambda}(x) | p, s \rangle. \quad (6.13)$$

The explicit factors of x^ν in $M^{\mu\nu\lambda}$ must be traded for $\partial/\partial k_\nu$ before the matrix element can be translated to $x = 0$:

$$\mathcal{M}^{\mu\nu\lambda}(p, k, s) = -i(2\pi)^4 \left[\frac{\partial}{\partial k_\nu} \{ \delta^4(p' + k - p) \langle p', s | T^{\mu\nu}(0) | p, s \rangle \} - (\lambda \leftrightarrow \nu) \right]. \quad (6.14)$$

Performing the k -differentiation and sending $k \rightarrow 0$, we find

$$\begin{aligned} \mathcal{M}^{\mu\nu\lambda}(p, 0, s) = -i(2\pi)^4 & \left[\langle p, s | T^{\mu\lambda}(0) | p, s \rangle \partial^\nu \delta^4(0) \right. \\ & \left. + \delta^4(0) \frac{\partial}{\partial k_\nu} \langle p', s | T^{\mu\lambda}(0) | p, s \rangle \right]_{k=0} - (\lambda \leftrightarrow \nu). \end{aligned} \quad (6.15)$$

Note the second term, which can be evaluated using the decomposition of eq. (6.9):

$$\begin{aligned} \mathcal{M}^{\mu\nu\lambda}(p, 0, s) = & \left[2p^\mu (p^\lambda i \partial^\nu - p^\nu i \partial^\lambda) \right. \\ & \left. + A_1(0) (2p^\mu \epsilon^{\lambda\nu\beta\sigma} - p^\nu \epsilon^{\mu\lambda\beta\sigma} - p^\lambda \epsilon^{\nu\mu\beta\sigma}) p_\beta s_\sigma \right] (2\pi)^4 \delta^4(0). \end{aligned} \quad (6.16)$$

The highly singular terms $\partial^\nu \delta^4(0)$ reflect the divergent integrals $\int d^4x x^\nu$. We have checked that if we had used wave packet states throughout there would have been no infinities and that the $\partial^\nu \delta^4(0)$ terms give rise to the contribution to $\mathcal{M}^{\mu\nu\lambda}$ from the center-of-mass motion of the wave packet.

Next we use angular momentum to normalize the s_σ -dependent term in eq. (6.16)*. We go to the rest frame and use

$$J^k \equiv \frac{1}{2} \epsilon^{kij} \int d^3x M^{0ij} \quad (6.17)$$

together with $J_3 |p = 0, \hat{s} = \hat{z}\rangle = \frac{1}{2} |p = 0, \hat{s} = \hat{z}\rangle$ and $s^2 = -M^2$ to obtain $A(0) = \frac{1}{2}$.

* The spin is defined in terms of the Pauli-Lubanski vector $W_\mu = \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} P^\nu J^{\rho\sigma}$, and is equal to eq. (6.17) in the rest frame. Note that P^ν and $J^{\rho\sigma}$ are gauge invariant and renormalization scale independent, as is W_μ .

So, finally, the forward matrix element of $M^{\mu\nu\lambda}$ in a spin- $\frac{1}{2}$ state is completely determined by four-momentum conservation (which fixed the norm of the term proportional to $\partial^\lambda \delta^4(0)$) and angular momentum conservation (which fixed the norm of the $\delta^4(0)$ term):

$$\mathcal{M}^{\mu\nu\lambda}(p, 0, s) = \left[2p^\mu(ip^\lambda \partial^\nu - ip^\nu \partial^\lambda) + \frac{1}{2}(2p^\mu \epsilon^{\lambda\nu\beta\sigma} - p^\nu \epsilon^{\mu\lambda\beta\sigma} - p^\lambda \epsilon^{\nu\mu\beta\sigma}) p_\beta s_\sigma \right] (2\pi)^4 \delta^4(0). \quad (6.18)$$

The sum rule for angular momentum can be written directly in terms of $\mathcal{M}^{\mu\nu\lambda}$. Let the nucleon be in an eigenstate of helicity $+\frac{1}{2}$ and choose its momentum to define the $\hat{e}_3$ -axis, then $p_0^\mu = (p^0, 0, 0, p^3)$ and $s_0^\mu = (p^3, 0, 0, p^0)$. Substituting in eq. (6.18) we obtain

$$\frac{\mathcal{M}^{012}(p_0, 0, s_0)}{(2\pi)^4 \delta^4(0) 2E} = \frac{1}{2}. \quad (6.19)$$

As soon as we have an expression for the tensor $M^{\mu\nu\lambda}$ in a model field theory, eq. (6.19) will allow us to construct the corresponding angular momentum sum rule in whatever reference frame we find convenient.

Finally, we explore an arbitrariness in $\mathcal{M}^{\mu\nu\lambda}$ and its implications for the angular momentum sum rule. In the context of some field theory let $B^{[\mu\beta][\nu\lambda]}$ be some operator antisymmetric in (μ, β) and in (ν, λ) . $B^{[\mu\beta][\nu\lambda]}$ may contain an explicit factor of the coordinate x . Then define a new tensor

$$M'^{\mu\nu\lambda} \equiv M^{\mu\nu\lambda} + \partial_\beta B^{[\mu\beta][\nu\lambda]}. \quad (6.20)$$

$M'^{\mu\nu\lambda}$ is conserved and antisymmetric in ν and λ . Note that the symmetry properties of M' (e.g. eq. (6.4)) may differ from those of M depending on the choice of B . The Lorentz generators defined from $M'^{\mu\nu\lambda}$ are the same as $J^{\nu\lambda}$:

$$J'^{\nu\lambda} = J^{\nu\lambda} + \int d^3x \partial_i B^{[0i][\nu\lambda]} = J^{\nu\lambda} \quad (6.21)$$

for field configurations which vanish at spatial infinity. Note also that the matrix elements of $M'^{\mu\nu\lambda}$ and $M^{\mu\nu\lambda}$ agree in the forward direction (∂_β generates an explicit multiplicative factor of q^β which cannot be cancelled in a theory with no massless physical states). Thus $B^{[\mu\beta][\nu\lambda]}$ represents an arbitrariness in $M^{\mu\nu\lambda}$ which cannot be eliminated by studying forward matrix elements and does not affect the angular momentum sum rule. The “superpotential” $B^{[\mu\beta][\nu\lambda]}$ will prove very handy in exposing the physical significance of the contributions to $M^{\mu\nu\lambda}$.

6.2. SPIN AND ORBITAL ANGULAR MOMENTUM OF QUARKS AND GLUONS

Although eq. (6.2) is a formally correct expression for $M^{\mu\nu\lambda}$, it fails to display the distinction between spin and orbital angular momentum expected on the basis of simple physical models. To make this separation in a meaningful way, we first use scalar field theory to define orbital angular momentum. For a free, real scalar

$$T_{(0)}^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - g^{\mu\nu} \mathcal{L}, \quad (6.22)$$

$$M_{(0)}^{\mu\nu\lambda} = \partial^\mu \phi (x^\nu \partial^\lambda - x^\lambda \partial^\nu) \phi - (g^{\mu\lambda} x^\nu - g^{\mu\nu} x^\lambda) \mathcal{L}. \quad (6.23)$$

Only the first term in $M^{\mu\nu\lambda}$ contributes to the angular momentum,

$$J^k \equiv \frac{1}{2} \epsilon^{ijk} J^{ij} = \int d^3x i \dot{\phi} \left(\mathbf{x} \times \frac{1}{i} \nabla \right)^k \phi. \quad (6.24)$$

Since $\dot{\phi} \equiv \pi$ is the momentum canonically conjugate to ϕ , this expression, evaluated in Fock space, would reduce to the sum over field modes of $\mathbf{r} \times \mathbf{p}$, which we consider a natural definition of orbital angular momentum. We generalize this to many component fields (of arbitrary spin) by summing over all canonical degrees of freedom:

$$L^k \equiv \int d^3x \sum_\alpha i \Pi^\alpha(x) \left(\mathbf{x} \times \frac{1}{i} \nabla \right)^k \Phi^\alpha(x), \quad (6.25)$$

where $(\Pi^\alpha, \Phi^\alpha)$ are pairs of canonically conjugate fields. Since constituent quark models and parton models are canonical, Fock-space models, this definition of L^k should correspond exactly to the intuitive definition of angular momentum in these models.

Consider next the case of a free, massive, spin- $\frac{1}{2}$ Dirac field, $\psi(x)$, where

$$T_{(1/2)\mu\nu} = \frac{1}{4} i \bar{\psi} (\gamma_\mu \partial_\nu + \gamma_\nu \partial_\mu) \psi + \text{h.c.}, \quad (6.26)$$

since $\mathcal{L} = 0$ by the equations of motion. $M^{\mu\nu\lambda}$ is

$$M_{(1/2)}^{\mu\nu\lambda} = \frac{1}{4} i \{ x^\nu \bar{\psi} (\gamma^\mu \partial^\lambda + \gamma^\lambda \partial^\mu) \psi - (\nu \leftrightarrow \lambda) \} + \text{h.c.} \quad (6.27)$$

$M_{(1/2)}^{\mu\nu\lambda}$ is conserved and has the proper symmetry [eq. (6.4)] but the $\gamma^\lambda \partial^\mu \psi$ term generates time derivatives not expected in J^k . A short calculation (using the Dirac equation) yields

$$\begin{aligned} M_{(1/2)}^{\mu\nu\lambda} = & \frac{1}{2} i \bar{\psi} \gamma^\mu (x^\nu \partial^\lambda - x^\lambda \partial^\nu) \psi + \text{h.c.} + \frac{1}{2} \epsilon^{\mu\nu\lambda\sigma} \bar{\psi} \gamma_\sigma \gamma_5 \psi \\ & - \frac{1}{8} \partial_\beta [\bar{\psi} x^\nu \{ \gamma^\lambda, \sigma^{\mu\beta} \} \psi - (\nu \leftrightarrow \lambda)]. \end{aligned} \quad (6.28)$$

The last term in this equation is of the form of a superpotential $\partial_\beta B^{[\mu\beta][\nu\lambda]}$, so we can define a new $M'^{\mu\nu\lambda}$ by dropping it:

$$M'_{(1/2)}{}^{\mu\nu\lambda} = \frac{1}{2}i\bar{\psi}\gamma^\mu(x^\nu\partial^\lambda - x^\lambda\partial^\nu)\psi + \frac{1}{2}\epsilon^{\mu\nu\lambda\sigma}\bar{\psi}\gamma_\sigma\gamma_5\psi. \quad (6.29)$$

From this we obtain a more familiar expression for the angular momentum in a spin- $\frac{1}{2}$ field theory:

$$J^k_{(1/2)} = \int d^3x i\psi^\dagger(\mathbf{x} \times \nabla)^k \psi + \frac{1}{2} \int d^3x \psi^\dagger \sigma^k \psi. \quad (6.30)$$

Since $i\psi^\dagger$ is the momentum canonically conjugate to ψ , the first term is an “orbital” angular momentum and the second term is the spin.

This familiar calculation has several interesting (and seductive) features: first, $\bar{\psi}\gamma_\sigma\gamma_5\psi \equiv A_\sigma^0$ is the flavor-singlet axial current, and second, it enters $M'_{(1/2)}{}^{\mu\nu\lambda}$ in the form $\frac{1}{2}\epsilon^{\mu\nu\lambda\sigma}A_\sigma^0$. Note however that by our general analysis, $M^{\mu\nu\lambda}$ has no totally antisymmetric part; furthermore the term, $\partial_\beta B^{[\mu\beta][\nu\lambda]}$ dropped from $M_{(1/2)}{}^{\mu\nu\lambda}$ to obtain $M'_{(1/2)}{}^{\mu\nu\lambda}$ has no forward matrix element. Thus, the forward matrix element of $M'_{(1/2)}{}^{\mu\nu\lambda}$ cannot have a term proportional to $\epsilon^{\mu\nu\lambda\sigma}$, the term of this form coming from $\frac{1}{2}\epsilon^{\mu\nu\lambda\sigma}A_\sigma^0$ must be exactly cancelled by a similar term coming from the “orbital” piece of $M_{(1/2)}{}^{\mu\nu\lambda}$.

Next, consider the case of a free, massless abelian gauge field, $A_\mu(x)$, where

$$T_{(g)}^{\mu\nu} = F^{\mu\alpha}F_\alpha^\nu - \frac{1}{4}g^{\mu\nu}F^2. \quad (6.31)$$

The generalizations to non-abelian gauge theories with or without matter fields are relatively straightforward and the essential difficulties are already present in the simplest case. $M^{\mu\nu\lambda}$ is

$$M_{(g)}^{\mu\nu\lambda} = x^\nu F^{\mu\alpha}F_\alpha^\lambda - x^\lambda F^{\mu\alpha}F_\alpha^\nu - \frac{1}{4}F^2(x^\nu g^{\mu\lambda} - x^\lambda g^{\mu\nu}). \quad (6.32)$$

Once again we seek a superpotential which will separate $M^{\mu\nu\lambda}$ into “orbital” and “spin” contributions. The momentum canonically conjugate to A^α is $F^{0\alpha}$, so the orbital term we seek is $F^{\mu\alpha}(-x^\nu\partial^\lambda + x^\lambda\partial^\nu)A_\alpha$. A little algebra yields

$$\begin{aligned} M_{(g)}^{\mu\nu\lambda} = & -F^{\mu\alpha}(x^\nu\partial^\lambda - x^\lambda\partial^\nu)A_\alpha + F^{\mu\lambda}A^\nu + F^{\nu\mu}A^\lambda - \frac{1}{4}F^2(x^\nu g^{\mu\lambda} - x^\lambda g^{\mu\nu}) \\ & + \partial_\beta(x^\nu F^{\mu\beta}A^\lambda - x^\lambda F^{\mu\beta}A^\nu). \end{aligned} \quad (6.33)$$

The first term is the orbital contribution, the second, $F^{\mu\lambda}A^\nu + F^{\nu\mu}A^\lambda$, is the spin, the third contributes only to boosts, and the fourth is a superpotential which we drop to obtain $M'_{(g)}{}^{\mu\nu\lambda}$. Integrating $M'_{(g)}{}^{0ij}$ we obtain an expression for the angular

momentum in an abelian gauge field theory:

$$J_{(g)}^k = \int d^3x E^l (\mathbf{x} \times \nabla)^k A^l + \int d^3x (\mathbf{A} \times \mathbf{E})^k. \quad (6.34)$$

Since E^l is the “momentum” conjugate to A^l , the “orbital” term has the expected form. The spin term is less familiar. If the first term is integrated by parts, we obtain

$$J_{(g)}^k = \int d^3x [\mathbf{x} \times (\mathbf{E} \times \mathbf{B})]^k \quad (6.35)$$

which may be the form most familiar to the reader.

Apparently, the gauge vector case is not the direct generalization of the Dirac case. First, the spin contribution to $M_{(g)}^{\mu\nu\lambda}$,

$$M_{(g)}^{\mu\nu\lambda}(\text{spin}) = F^{\mu\lambda}A^\nu + F^{\nu\mu}A^\lambda \quad (6.36)$$

is not directly related to the U(1) axial current, K_σ . The analogy with the Dirac case would have led one to expect $M_{(g)}^{\mu\nu\lambda}(\text{spin})$ to be

$$\epsilon^{\mu\nu\lambda\sigma}K_\sigma = F^{\mu\lambda}A^\nu + F^{\nu\mu}A^\lambda + F^{\lambda\nu}A^\mu, \quad (6.37)$$

but the term $F^{\lambda\nu}A^\mu$ is missing from $M_{(g)}^{\mu\nu\lambda}(\text{spin})$. Furthermore, $F^{\lambda\nu}A^\mu$ is not separately conserved so there is no superpotential which can convert the spin term to the form (6.37). $\epsilon^{\mu\nu\lambda\sigma}K_\sigma$ and $M_{(g)}^{\mu\nu\lambda}(\text{spin})$ are not in the same representation of the Lorentz group. The former is a vector $(\frac{1}{2}, \frac{1}{2})$ while the latter contains $(\frac{1}{2}, \frac{3}{2}) \oplus (\frac{3}{2}, \frac{1}{2})$ in addition to $(\frac{1}{2}, \frac{1}{2})$. Finally, the orbital and spin terms in $M_{(g)}^{\mu\nu\lambda}$ are not separately gauge invariant (although $M_{(g)}^{\mu\nu\lambda}$ is gauge invariant up to a superpotential). So the decomposition into spin and orbital parts is meaningful only with reference to a specific gauge.

The distinction between $M_{(g)}^{\mu\nu\lambda}(\text{spin})$ and $\epsilon^{\mu\nu\lambda\sigma}K_\sigma$ becomes less clear when one constructs the rotation generators. The spin contribution to $J_{(g)}$ is $\int d^3x (\mathbf{A} \times \mathbf{E})$. The analogous integral over $\epsilon^{\mu\nu\lambda\sigma}K_\sigma$ contains an additional term: $\int d^3x A^0 \mathbf{B}$. In $A^0 = 0$ gauge the two expressions agree. The same is true for the analogous generator ($\int d^2x_\perp dx^- M^{+ij}$) for a theory quantized on the light-cone in $A^+ = 0$ gauge.

Finally, we quote the generalization of $M^{\mu\nu\lambda}$ to QCD. With a suitable choice of a superpotential we obtain

$$\begin{aligned} M_{\text{QCD}}^{\mu\nu\lambda} = & \frac{1}{2} i \bar{\psi} \gamma^\mu (x^\lambda \partial^\nu - x^\nu \partial^\lambda) \psi + \frac{1}{2} \epsilon^{\mu\nu\lambda\sigma} \bar{\psi} \gamma_\sigma \gamma_5 \psi - 2 \text{Tr} \{ F^{\mu\alpha} (x^\nu \partial^\lambda - x^\lambda \partial^\nu) A_\alpha \} \\ & + 2 \text{Tr} \{ F^{\mu\lambda} A^\nu + F^{\nu\mu} A^\lambda \} - \frac{1}{2} \text{Tr} F^2 (x^\nu g^{\mu\lambda} - x^\lambda g^{\mu\nu}). \end{aligned} \quad (6.38)$$

The terms in eq. (6.38) correspond successively to the orbital and spin angular momentum of the quarks and gluons. Note that with the exception of $\epsilon^{\mu\nu\lambda\sigma}\bar{\psi}\gamma_\sigma\gamma_5\psi$, the terms in (6.38) are not separately gauge invariant, and mix under renormalization. In the non-abelian case, $M_{(g)}^{\mu\nu\lambda}$ (spin) does not have the $F^{\lambda\nu}A^\mu$ term (as in the abelian case), nor does it have the $(A)^3$ terms in $\epsilon^{\mu\nu\lambda\sigma}K_\sigma$.

6.3. CONSTRUCTING AND INTERPRETING SUM RULES

With an eye toward making contact with the parton model we construct a sum rule from eq. (6.19) in an “infinite momentum” frame, $p^3 \equiv p_\infty \rightarrow \infty$. [We could just as well use light-cone variables, M^{+ij} etc., in the rest frame.]

$$\begin{aligned} \frac{1}{2} = & \left\langle p_\infty^0, s^0 \left| \int d^3x \left[i\psi^\dagger (\mathbf{x} \times \nabla)^3 \psi + \frac{1}{2} \psi^\dagger \sigma^3 \psi \right. \right. \right. \\ & \left. \left. + 2 \text{Tr} \{ E^k (\mathbf{x} \times \nabla)^3 A^k + (\mathbf{A} \times \mathbf{E})^3 \} \right] \right| p_\infty^0, s^0 \right\rangle / 2E(2\pi)^3 \delta^3(0). \quad (6.39) \end{aligned}$$

If there is any problem with the convergence of the x -integration in the orbital terms (containing explicit factors of x^k), the matrix element should be interpreted as the limit of an expectation value in a wave packet with nearly sharp momentum. Each term in eq. (6.39) is renormalization scale dependent and should carry a label μ^2 to indicate the renormalization point. Here and subsequently we suppress this label.

The quark and gluon spin terms are local so we may factor out the normalization $2E(2\pi)^3 \delta^3(\mathbf{0})$ and define

$$\begin{aligned} \Sigma &= \langle p_\infty^0, s^0 | \psi^\dagger \sigma^3 \psi | p_\infty^0, s^0 \rangle \\ &= \langle p_\infty^0, s^0 | \bar{\psi} \gamma^3 \gamma_5 \psi | p_\infty^0, s^0 \rangle, \\ \Delta g &= \langle p_\infty^0, s^0 | 2 \text{Tr} (\mathbf{A} \times \mathbf{E})^3 | p_\infty^0, s^0 \rangle, \quad (6.40) \end{aligned}$$

which relate the quark and gluon spin contribution to the nucleon's angular momentum to the matrix elements of local operators.

If we choose a gauge in which there are no unphysical gauge degrees of freedom (e.g. $A^0 = 0$ gauge), we can define independent expressions for Σ and Δg in terms of the distribution of quarks and gluons with helicity parallel and antiparallel to the nucleon's in the infinite momentum frame:

$$\Sigma = \int_0^1 dx (q_\uparrow(x) - q_\downarrow(x)), \quad (6.41)$$

$$\Delta g = \int_0^1 dx (g_\uparrow(x) - g_\downarrow(x)). \quad (6.42)$$

[A sum over quarks and antiquarks of all flavors is understood, as is the renormalization scale dependence of all terms.] The first sum rule,

$$\int_0^1 dx (q_\uparrow(x) - q_\downarrow(x)) = \left\langle p, \lambda = \frac{1}{2} \left| \bar{\psi} \gamma^3 \gamma_5 \psi \right| p, \lambda = \frac{1}{2} \right\rangle, \quad (6.43)$$

is the familiar quark spin sum rule. The second,

$$\int_0^1 dx (g_\uparrow(x) - g_\downarrow(x)) = \left\langle p, \lambda = \frac{1}{2} \left| (A \times E)^3 \right| p, \lambda = \frac{1}{2} \right\rangle, \quad (6.44)$$

is the analogous result for gluons (in $A^0 = 0$ gauge). In this gauge, the operator $(A \times E)^3$ coincides with K^3 , so in this qualified sense the axial gauge current is a measure of the gluon spin contribution to the nucleon's spin. However, gauge dependence of the forward matrix element of K_μ discussed in sect. 5 limits usefulness of this identification.

Combining these results we obtain the “angular momentum” sum rule

$$\Delta L_q + \frac{1}{2} \Sigma + \Delta L_g + \Delta g = \frac{1}{2}, \quad (6.45)$$

where

$$\Delta L_q = \frac{1}{2E(2\pi)^3 \delta^3(0)} \left\langle p_\infty^0, s^0 \left| \int d^3x i \psi^\dagger (\mathbf{x} \times \nabla)^3 \psi \right| p_\infty^0, s^0 \right\rangle, \quad (6.46)$$

$$\Delta L_g = \frac{1}{2E(2\pi)^3 \delta^3(0)} \left\langle p_\infty^0, s^0 \left| \int d^3x \text{Tr} \{ E^k (\mathbf{x} \times \nabla)^3 A^k \} \right| p_\infty^0, s^0 \right\rangle. \quad (6.47)$$

Σ is measurable, but as far as we know the other three terms are not, nor are they separately gauge or renormalization invariant.

Several recent theoretical studies of the spin content of the nucleon [11,14] begin by writing a “parton model” sum rule like eq. (6.45) but omitting the quark and gluon orbital angular momentum:

$$\frac{1}{2} \Sigma + \Delta g \stackrel{?}{=} \frac{1}{2}. \quad (6.48)$$

This sum rule is false. There is no reasonable approximation to the parton model in which eq. (6.48) is valid. The error in omitting ΔL_g and ΔL_q is of exactly the same order as the error in estimating Σ by unity as discussed in sect. 3. The quark and gluon orbital angular momentum are naively of the same order as Σ and Γ . Partons moving with a fraction x of the nucleon's (“infinite”) momentum may nevertheless have small transverse momentum components, $\mathbf{k}_\perp$. $\mathbf{k}_\perp$ can be shown to contribute only to $O(k_\perp^2/Q^2)$ (i.e. higher twist) corrections to deep inelastic

phenomena, and is often ignored in leading twist calculations. This is not the case insofar as the component of angular momentum along $\mathbf{p}$ is concerned. A bound state with arbitrarily soft internal momentum can have arbitrary orbital angular momentum because the product kR achieves a finite limit in a given partial wave as $k \rightarrow 0$.

To make this explicit we have constructed an example. Consider a Dirac particle confined by a potential $U(\rho)$ in the transverse plane. Since J_3 and p_3 are constants of the motion, it is possible to construct an energy eigenstate of very large (“infinite”) momentum, k , in the 3-direction, with arbitrary angular momentum along the $\hat{e}_3$ -axis, j_3 . The Dirac wave function (with $\beta = \rho^3$, $\gamma = i\rho^2\sigma$) is

$$\psi(\rho, \phi, z) = \begin{pmatrix} f_1(\rho) + \sigma \cdot \hat{\rho} g_1(\rho) \\ \sigma \cdot \hat{\rho} f_2(\rho) + g_2(\rho) \end{pmatrix} e^{in\phi} e^{ikz} u_{\pm}, \quad (6.49)$$

with $u_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $u_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $j_3 = n \pm \frac{1}{2}$. The functions f_i and g_i are determined by

$$\begin{aligned} i \left(-f_1' + \frac{n}{\rho} f_1 \right) &= (E + m + U(\rho)) f_2 + k g_1, \\ i \left(-f_2' - \frac{n+1}{\rho} f_2 \right) &= (E - m - U(\rho)) f_1 - k g_2, \\ i \left(-g_1' - \frac{n+1}{\rho} g_1 \right) &= (E + m + U(\rho)) g_2 - k f_1, \\ i \left(-g_2' + \frac{n}{\rho} g_2 \right) &= (E - m - U(\rho)) g_1 + k f_2. \end{aligned} \quad (6.50)$$

These equations have two linearly independent solutions (regular at $\rho = 0$) for each choice of n corresponding to the two solutions of opposite parity for each j in the spherically symmetric case.

Even a potential of modest depth and range will bind several partial waves leading to several values of j_3 . Clearly, orbital angular momentum plays a crucial role even if $j_3 = \pm \frac{1}{2}$. To be quantitative we calculate $\langle \sigma_3 \rangle$:

$$\langle \sigma_3 \rangle = \pm \frac{\int_0^\infty \rho d\rho (f_1^2 + g_2^2 - f_2^2 - g_1^2)}{\int_0^\infty \rho d\rho (f_1^2 + g_2^2 + f_2^2 + g_1^2)},$$

where the overall sign depends on the choice of the spinor $u_{\pm}$. For relativistic quarks all four functions, f_1 , f_2 , g_1 and g_2 are comparable. Thus even for $j_3 = \pm \frac{1}{2}$ the quark spin may be completely *anticorrelated* with its total angular momentum.

This simply reconfirms the argument from sect. 3 that the upper and lower components of the Dirac wave function in a spherical potential have opposite spin orientation on the average. As discussed in sect. 3, exactly this effect is responsible for decreasing g_A/g_V from 5/3 to 1.25 in relativistic quark models, so we have experimental evidence (in the context of a quark model) that the orbital angular momentum of quarks cannot be ignored.

7. Calculation of Σ in the Skyrme model

In the Skyrme model, the quark spin operator may be identified with the flavor-singlet axial vector current. In ref. [9], it was shown that $\Sigma = 0$ in the $N_c \rightarrow \infty$ and SU(3) symmetry limits of the three-flavor Skyrme model. Including SU(3) symmetry violation, the result becomes $\Sigma = -0.18$. This is an interesting calculation, but must be taken with a grain of salt for the following reasons.

(i) The Skyrme model is supposed to describe QCD in the large N_c limit. $1/N_c$ effects are extremely large in the $U(1)_A$ sector. For example, $m_{\eta'}/m_\rho \propto N_c^{-3/2}$, but in the real world $m_{\eta'}/m_\rho = 1.02$. The leading term in the $1/N_c$ expansion is likely to be unreliable in the $U(1)_A$ sector.

(ii) The approximation used in ref. [9] implies a stable η' . If the Skyrme lagrangian is modified to include the lowest dimension operators responsible for $\eta' \rightarrow \eta\pi\pi$, then one obtains [25] $\Sigma = 0.2 \pm 0.1$.

(iii) The same limit ($N_c \rightarrow \infty$) of the SU(3) symmetric Skyrme model fails to describe the flavor pattern of other nucleon matrix elements. Recently, Kaplan and Klebanov [26] re-examined Donoghue and Nappi's [56] calculation of the matrix elements of the scalar operators, $\bar{q}q$. Donoghue and Nappi fixed $N_c = 3$ and found $2\langle \bar{s}s \rangle / [\langle \bar{u}u + \bar{d}d \rangle]$ of order unity as apparently required by experiment (see sect. 8). Kaplan and Klebanov pointed out that

$$\frac{2\langle \bar{s}s \rangle}{\langle \bar{n}n + \bar{d}d \rangle} \sim \frac{2}{N_c} + O\left(\frac{1}{N_c^2}\right), \quad (7.1)$$

so $\langle \bar{s}s \rangle \approx 0$ in the same limit ($N_c \rightarrow \infty$) that $\Sigma \approx 0$. One response to this would be to forget $N_c \rightarrow \infty$ and embrace the Donoghue–Nappi calculation, but this leaves the desired result $\Sigma \approx 0$ without justification. Kaplan and Klebanov suggest several more imaginative ways to study flavor structure in the Skyrme model.

(iv) If the octet axial vector charges calculated in the SU(3) Skyrme model [57] are combined with the result for Σ , the results for Δu , Δd and Δs are not impressive when compared with the “EMC” values, eq. (2.17).

$$\Delta u_{\text{Skyrme}} \cong 0.50\text{--}0.55, \quad \Delta d_{\text{Skyrme}} \cong -0.38\text{--}-0.30, \quad \Delta s_{\text{Skyrme}} \cong -0.12\text{--}0.04,$$

where the first number is the value for $\Sigma = 0$, and the second for $\Sigma = 0.2$.

(v) Since the SU(2) Skyrme model describes a proton without strange quark content, claiming $\Delta s \neq 0$ to be a success of the SU(3) Skyrme model requires one to re-examine the traditional successes of the SU(2) Skyrme model.

8. The flavor structure of the nucleon matrix elements of quark operators

In sect. 3 we argued that a small value of $\Sigma(Q_{\text{EMC}}^2)$ could be accommodated in a fairly natural way in both non-relativistic quark models and bag models, provided one is not overly naive about the formulation of such models. We do not mean to suggest that there is nothing perplexing about the EMC result. In this section we contrast the axial vector data with measurements of nucleon forward matrix elements of other quark bilinear operators. The overall situation is confusing: some strange-quark matrix elements are large, others, among them the most precisely measured, are very small. Here we briefly summarize what we know about the nucleon matrix elements of bilinear quark operators and mention some recent proposals for new experiments. We have not attempted to be complete: some of the topics we include, notably the πN sigma term, would require a monograph at least as long as this one to handle them adequately.

The operators we consider are of the term $\bar{\psi} \Gamma D^{\mu_1} \dots D^{\mu_n} \psi$, where Γ is a Dirac γ -matrix. Some of these operators are renormalization scale invariant, others are not and any measurement and/or comparison with theory must be referred to a definite mass scale. In all cases we are particularly interested in the relative magnitude of the strange and non-strange matrix elements.

8.1. VECTOR CHARGES

As a warm-up consider

$$V_\mu^a = \bar{\psi} \gamma_\mu T^a \psi, \quad a = 1, \dots, 8 \quad (8.1)$$

$$V_\mu = \bar{\psi} \gamma_\mu \psi. \quad (8.2)$$

All non-vanishing nucleon matrix elements of these operators are determined by conserved quantum numbers. Since the nucleon has baryon number and hypercharge one, we find

$$\begin{aligned} \langle p, s | \bar{s} \gamma_\mu s | p, s \rangle &= 0, \\ \langle p, s | \bar{u} \gamma_\mu u + \bar{d} \gamma_\mu d | p, s \rangle &= 3(2p_\mu), \end{aligned} \quad (8.3)$$

which simply counts quarks minus antiquarks of different flavors.

8.2. AXIAL CHARGES

The octet and singlet axial charges are defined by eqs. (2.6) and (2.12). The EMC data (corroborated by $\nu p \rightarrow \nu p$) determine the strange and non-strange matrix elements at $Q_{\text{EMC}}^2 \cong 10 \text{ GeV}^2$:

$$\langle p, s | \bar{s} \gamma_\mu \gamma_5 s | Q^2 | p, s \rangle = \Delta s(Q^2) s_\mu, \quad (8.4)$$

$$\langle p, s | \bar{u} \gamma_\mu \gamma_5 u + \bar{d} \gamma_\mu \gamma_5 d | Q^2 | p, s \rangle = (\Delta u(Q^2) + \Delta d(Q^2)) s_\mu, \quad (8.5)$$

with

$$\Delta s(Q_{\text{EMC}}^2) = -0.16 \pm 0.08,$$

$$\Delta u(Q_{\text{EMC}}^2) + \Delta d(Q_{\text{EMC}}^2) = 0.28 \pm 0.13, \quad (8.6)$$

indicating a significant component of polarized strange quarks in the nucleon.

8.3. TWIST-2 VECTOR AND AXIAL VECTOR OPERATORS IN LEPTON SCATTERING

We define $V_n(\psi)$ and $A_n(\psi)$ by

$$S\left(\frac{i}{2}\right)^{n-1} \langle p, s | \bar{\psi} \gamma^{\mu_1} \vec{D}^{\mu_2} \dots \vec{D}^{\mu_n} \psi | p, s \rangle \equiv 2V_n(\psi) S\{p^{\mu_1} \dots p^{\mu_n}\}, \quad (8.7)$$

$$S\left(\frac{i}{2}\right)^{n-1} \langle p, s | \bar{\psi} \gamma^{\mu_1} \vec{D}^{\mu_2} \dots \vec{D}^{\mu_n} \gamma_5 \psi | p, s \rangle \equiv 2A_n(\psi) S\{p^{\mu_1} p^{\mu_2} \dots p^{\mu_n}\}, \quad (8.8)$$

where ψ can be either u , d or s . V_n for n even and A_n for n odd are related to the structure functions in deep inelastic ep scattering. V_1 and A_1 are, of course, the vector and axial charges discussed above. The $\{V_n(\psi)\}$ and $\{A_n(\psi)\}$ depend in a complicated way on the renormalization scale, μ^2 : the non-singlet parts evolve logarithmically at large μ^2 , the singlet parts mix with gluonic operators of the same twist and spin (except, of course, for A_1). For simplicity, we suppress the μ^2 label. Since we are interested only in isospin invariant ratio $\langle \bar{s}s \rangle / \langle \bar{u}u + \bar{d}d \rangle$ we can consider data on proton and nuclear targets interchangeably (ignoring the small corrections from the “old” EMC effect).

The operator $V_2(\psi)$ is particularly interesting, first because it has the parton interpretation of the fraction of the nucleon momentum carried by quark- ψ , and second because it is rather precisely measured for strange and non-strange quarks. Strange quarks can be separated from up and down quarks by measuring charm production in neutrino scattering. The idea, illustrated in fig. 5, is that neutrino scattering from an s-quark produces ($\cos \theta_c \approx 1$) a leading charm quark which decays $\sim 20\%$ of the time to $\mu^+ \bar{\nu}_\mu X$ producing a leading μ^+ along q . [In fact, the

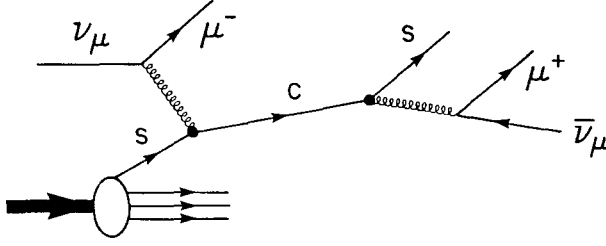


Fig. 5. Parton model picture of charm production in neutrino-hadron scattering.

excess of opposite sign dimuons in ν -scattering was a puzzle resolved by the discovery of charm.] In this way CDHS [20] has measured that $2.6 \pm 0.6\%$ of the nucleon's momentum is carried by s and $\bar{s}$ quarks. $\bar{u}$ and $\bar{d}$ quarks together carry twice as much momentum. All together, u , d , $\bar{u}$, and $\bar{d}$ quarks carry $43.8 \pm 0.2\%$ of the nucleon's momentum. These numbers have been confirmed by the CCFRR [20] collaboration:

$$\begin{aligned} V_2(s) &= 0.026 \pm 0.006, \\ V_2(u) + V_2(d) &= 0.438 \pm 0.002, \end{aligned} \quad (8.9)$$

at the (large) momentum scale of these experiments.

These measurements are quite accurate (by the standards of this field) and support a picture of a nucleon whose "intrinsic" wave function is devoid of s (and $\bar{s}$) quarks, and which develops a small $s\bar{s}$ -component by means of the perturbative evolution from the intrinsic scale (≤ 1 GeV) to the Q^2 at which the experiments are performed. Indeed, the measurements indicate that the strange "sea" in the nucleon is suppressed by about a factor two compared to the non-strange pair sea:

$$\frac{\langle s + \bar{s} \rangle}{\langle \bar{u} + \bar{d} \rangle} = 0.52 \pm 0.09. \quad (8.10)$$

By considering imaginary worlds where the quark charges are different from their present ones, and requiring that the ep scattering cross section be positive, i.e. by unitarity, we are able to derive inequalities on the A_n and V_n [58]:

$$|A_1(\psi)| \geq |A_3(\psi)| \geq |A_5(\psi)| \geq \dots, \quad (8.11)$$

$$V_2(\psi) \geq V_4(\psi) \geq \dots \geq 0, \quad (8.12)$$

$$V_n(\psi) \geq |A_{n+1}(\psi)|. \quad (8.13)$$

The inequalities hold separately for each possible flavor of $\psi(u, d, s)$. The inequalities (8.11)–(8.13) immediately imply that

$$V_{2n}(s) \leq 0.026, \quad n \geq 1, \quad (8.14)$$

$$|A_{2n+1}(s)| \leq 0.026, \quad n \geq 1, \quad (8.15)$$

i.e. the only possibly large strange quark matrix element in the proton among the $\{V_n\}$ and $\{A_n\}$ is A_1 , the matrix element of $\bar{s}\gamma^\mu\gamma_5 s$.

8.4. SCALAR DENSITIES

The scalar densities, $\bar{\psi}\psi$, are not renormalization point invariant, but they scale like inverse quark masses, so the products $m_\psi\bar{\psi}\psi$ which are chiral symmetry violating terms in the QCD hamiltonian are renormalization scale invariant. The nucleon matrix element of the operator

$$\Sigma \equiv \bar{m}(\bar{u}u + \bar{d}d) \quad (8.16)$$

($\bar{m} = m_u + m_d$) can be extracted from a complicated analysis of low-energy πN scattering [59]. After years of effort its value has converged toward

$$\bar{m}\langle p, s | (\bar{u}u + \bar{d}d) | p, s \rangle \cong 60 \pm 5 \text{ MeV}, \quad (8.17)$$

although this value is not universally accepted [60]. On the other hand, the operator

$$H_{\text{SB}}(\text{SU}(3)) = \frac{\bar{m} - m_s}{3}(\bar{u}u + \bar{d}d - 2\bar{s}s) \quad (8.18)$$

is the term in the QCD hamiltonian which violates SU(3) symmetry. If its matrix elements can be reliably extracted from baryon masses using first-order perturbation theory, then the results can be combined with the previous result, and quark mass ratios obtained from the pseudoscalar meson sector using chiral SU(3) $\times$ SU(3) to obtain

$$m_s\langle p, s | \bar{s}s | p, s \rangle = 330 \pm 130 \text{ MeV}, \quad (8.19)$$

$$\frac{\langle p, s | \bar{s}s | p, s \rangle}{\frac{1}{2}\langle p, s | \bar{u}u + \bar{d}d | p, s \rangle} \cong \frac{1}{2}. \quad (8.20)$$

How surprising is this result? In a naive bag model or non-relativistic quark model in which the proton contains no strange quark pairs, it appears very strange: the numerator receives contributions only from the $q\bar{q}$ -sea, whereas the denominator receives “valence” contributions. Less naive models like chiral quark models [38]

and hybrid bag models [58] suggest a more complicated situation. In chiral quark models the SU(3) singlet scalar operators and octet may be renormalized differently in the transition from current to constituent quarks [5]:

$$U^0 = \bar{u}u + \bar{d}d + \bar{s}s = g_s^{(0)}(\bar{U}U + \bar{D}D + \bar{S}S), \quad (8.21)$$

$$U^a = \bar{\psi}T^a\psi = g_s^{(8)}(\bar{\Psi}T^a\Psi). \quad (8.22)$$

The OZI rule requires $g_s^{(0)} = g_s^{(8)}$ and $\langle p, s | \bar{s}s | p, s \rangle = 0$. Mixing in the singlet channel frees $g_s^{(0)}$ from $g_s^{(8)}$ and can accommodate the ratio eq. (8.20). The only defect we see in this point of view is the striking success of the OZI rule in the case of the “momentum sum rule,” $V_2(\psi)$.

In hybrid bag models [61] and in various toy models of dynamical chiral symmetry violation, eq. (8.20) is avoided because the use of first-order perturbation theory to evaluate the matrix elements of $H_{\text{SB}}(\text{SU}(3))$ is not valid. In these models eq. (8.17) is compatible with $\langle p, s | \bar{s}s | p, s \rangle \ll \frac{1}{2} \langle p, s | \bar{u}u + \bar{d}d | p, s \rangle$.

8.5. MAGNETIC MOMENTS

Quark magnetic moment operators can be defined in analogy to the previous cases

$$\boldsymbol{\mu}(\psi) \equiv \frac{1}{2} \mathbf{x} \times \bar{\psi} \boldsymbol{\gamma} \psi \quad (8.23)$$

for each flavor $\psi = u, d, s$. The proton matrix elements of these operators are related to the standard F_1 and F_2 form factors for the vector currents V_μ :

$$\langle p = 0, s | \boldsymbol{\mu}(\psi) | p = 0, s \rangle \equiv \mu_N(\psi) \mathbf{s} = [F_{1\psi}(0) + F_{2\psi}(0)] \mathbf{s}, \quad (8.24)$$

where

$$\langle p + q, s | \bar{\psi} \boldsymbol{\gamma}_\mu \psi | p, s \rangle = \bar{U}(p + q, s') \left\{ F_{1\psi}(q^2) \boldsymbol{\gamma}_\mu + F_{2\psi}(q^2) \frac{i \boldsymbol{\sigma}_{\mu\nu} q^\nu}{2M_N} \right\} U(p, s). \quad (8.25)$$

The proton's quantum numbers require

$$F_{1u}(0) = 2, \quad F_{1d}(0) = 1, \quad F_{1s}(0) = 0. \quad (8.26)$$

The proton and neutron electromagnetic moments determine two linear combinations of F_{2u} , F_{2d} and F_{2s} ,

$$F_{2u} - F_{2d} = \kappa_p - \kappa_n = 3.71,$$

$$F_{2u} + F_{2d} - 2F_{2s} = 3(\kappa_p + \kappa_n) = -0.36,$$

but do not provide any information about the flavor singlet combination $F_2^{(0)} \equiv \frac{1}{3}(F_{2u} + F_{2d} + F_{2s})$ [5]. $F_2^{(0)}$ can be measured in elastic νp scattering [5], or, as pointed out by McKeown [22] and discussed in further detail by Beck [23], by a precise measurement of the parity-violating asymmetry in elastic ep scattering. Until such time as these experiments are performed, the prediction of $F_2^{(0)}$ remains an excellent goal for theorists who think they have understood the flavor structure of the nucleon. To begin the debate we record the predictions for $F_2^{(0)}$ in some of the models we have discussed in this paper.

(i) Non-relativistic quark model in which the current and constituent quarks are identified + EMC data. [The magnetic moment operator and the axial charge are taken as proportional to each other.] $F_2^{(0)} = \frac{1}{3}\Sigma - 1 \approx -1$.

(ii) "Naive" quark model (no s-quarks in the nucleon): $F_2^{(0)} = \frac{1}{3}\mu_p - 1 = -0.12$.

(iii) Chiral quark model [5]: $F_2^{(0)} = -0.12 \pm 0.2 \times O(1)$.

(iv) Two-flavor Skyrme model in the SU(2) limit: $F_2^{(0)} = -0.45$. (The calculation for the three-flavor model is in progress.)

(v) Recently one of us has attempted to extract $F_2^{(0)}$ from data on the nucleon isoscalar form factor and information about the flavor structure of vector mesons [62] with the result $F_2^{(0)} \approx -0.43$.

The predictions vary significantly enough to be distinguished by a relatively imprecise measurement of $F_2^{(0)}$.

8.6. CHARGE RADII, ETC.

Finally, we mention the higher (coordinate space) moments of the vector and axial currents which can be extracted from the study of elastic electron and quasielastic neutrino scattering from proton and neutron targets. Specific quark operators can be identified with the first non-trivial q^2 -derivatives of the various form factors which appear in the slightly off-diagonal matrix elements of these currents. The methods of ref. [62] gives $\langle r_s^2 \rangle \approx 0.16 \text{ fm}^2$ compared to $\langle r_0^2 \rangle = 1.06 \text{ fm}^2$. At the present time only a few of these charge radii are well measured (e.g. electric and magnetic charge radius of the proton and neutron). In some cases uncertainty about the relevant charge radii (i.e. the mass scale in the form factor) complicates the extraction of the charge itself from data. An example is the ambiguity noted by Kaplan and Manohar (and by the experimenters themselves) in the $\nu p \rightarrow \nu p$ data affecting the extraction of Σ . In this particular case a new experiment being planned at LAMPF hopes to eliminate the ambiguity by limiting itself to extremely low momentum transfers [37].

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